

Research on the acceptance of Chinese traditional music and student cultural identity differences under Chinese and Western music education models

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ABSTRACT

In the era of globalization and cultural diversification, the collision and integration of Chinese and Western music education models have profoundly influenced students' cultural identity formation. This study aims to explore the influencing mechanism of different music education models on the acceptance of Chinese traditional music and the mediating role of cultural identity. Using stratified random sampling, 1156 students (46.3% male, aged 12–22) from eastern (36.8%), central (33.2%), and western (30.0%) regions who received different music education models were surveyed. Theoretical hypotheses were tested through structural equation modeling, mediation effect analysis, and multi-group comparison methods. Comprehensive assumption testing confirmed data suitability for SEM analysis (all skewness |values| < 2, kurtosis |values| < 7, VIF range: 1.23–2.87), and measurement invariance was established across education model groups (ΔCFI values < 0.01 for configural, metric, and scalar invariance). The findings reveal that: (1) students primarily receiving traditional music education show significantly higher acceptance ($M = 5.24$, $SD = 0.98$) compared to comprehensive ($M = 4.58$, $SD = 1.12$) and Western music education models ($M = 3.87$, $SD = 1.35$), $F(2,1153) = 78.42$, $p < 0.001$, $\eta^2 = 0.12$; (2) cultural identity plays a partial mediating role ($\beta = -0.198$, $p < 0.001$), accounting for 47.7% of the total effect; (3) family music resources significantly moderate the impact ($\beta = 0.142$, $p < 0.001$); (4) western regions show higher transformation efficiency ($\beta = 0.61$) than eastern regions ($\beta = 0.48$), $\Delta\chi^2 = 9.87$, $p = 0.007$; (5) music participation positively moderates the mediating effect (Index = 0.082, 95% CI [0.041, 0.123]). The model explains 23.8% of variance in cultural identity and 41.7% in traditional music acceptance. Model robustness was confirmed through alternative model comparisons, cross-validation (RMSE difference < 8%), and sensitivity analyses. However, the cross-sectional design limits causal inference, and future longitudinal research is needed to establish temporal precedence. The results indicate that music education models influence traditional music acceptance through dual pathways of direct transmission and cultural shaping, providing theoretical support and practical guidance for constructing a Chinese-Western integrated music education system while maintaining cultural subjectivity.

1. Introduction

In the era of globalization and cultural diversification, music education, as an important carrier of cultural inheritance and identity construction, is experiencing profound changes and reflections in its educational models and practical approaches. Chinese traditional music, as an important component of Chinese culture, carries profound historical heritage and unique aesthetic value systems. However, in the contemporary educational environment, the introduction of Western music education models and their collision with local traditional music

education systems have produced complex issues of cultural adaptation and identity. This cross-cultural educational practice not only affects students' perception and acceptance of traditional music but also more deeply shapes the younger generation's cultural identity formation and value concepts.

Current international academic research has conducted multi-dimensional explorations of cultural identity issues in music education, with research perspectives covering the impact of cross-cultural music participation on social cohesion and the shaping effect of education models on student cultural identity. Crooke et al. (2024) have

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constructed a conceptual framework for cross-cultural music participation. This framework explains how music promotes understanding between groups through multimodal activity. The study reveals that music facilitates integration through cognitive, emotional, social, and physical participation dimensions (Crooke et al., 2024). Ford's research (2020) deeply analyzes cultural assimilation in Western classical music education. The study reveals implicit cultural biases faced by Asian students. It also identifies identity dilemmas in Western music education systems (Ford, 2020). Wang and Webb (2023) found multiple challenges through systematic literature review. These challenges include teaching methods and cultural understanding barriers. Teacher professional development emerges as another critical challenge (Wang & Webb, 2023). The study emphasizes the necessity of culturally sensitive teaching models (Wang & Webb, 2023). Additionally, empirical research demonstrates the effectiveness of cultural identity affirmation activities. Such activities significantly enhance students' recognition of their cultural background in secondary education (Thompson & Garcia, 2021). However, teachers' cultural awareness plays a key role in this process. Teaching strategies directly impact the effectiveness of cultural identity formation (Thompson & Garcia, 2021). Recent technological advances also influence cross-cultural music education. Studies explore synchronous online music teaching in cross-cultural environments (Roberts & Kim, 2023). These innovations present both opportunities and practical challenges (Roberts & Kim, 2023).

Based on the theoretical foundation and practical exploration of existing research, this study innovatively combines comparative analysis of Chinese and Western music education models with research on student cultural identity differences. The research constructs multi-dimensional theoretical frameworks through empirical models. It systematically explores influencing mechanisms of traditional music acceptance. The study examines pathways in shaping student cultural identity formation under different education models. The research not only focuses on characteristic differences of education models themselves but also deeply explores the mechanisms of moderating factors such as family background and regional culture in this process. Through quantitative analysis methods such as structural equation modeling, it reveals the mediating effect of cultural identity between music education and traditional music acceptance, providing theoretical support and empirical evidence for optimizing music education practice and promoting cultural inheritance and innovative development. The theoretical framework constructed in this study is shown in Fig. 1, which integrates core elements such as education model characteristics, cultural identity dimensions, traditional music acceptance, and moderating variables, clearly presenting the logical relationships and mechanisms

between variables.

2. Literature review and theoretical framework

2.1. Historical evolution and characteristic comparison of Chinese and Western music education models

The developmental evolution of music education models profoundly reflects the historical changes in cultural values and educational philosophies. The Western music education system has gradually formed a teaching paradigm centered on individual creativity, critical thinking, and technical precision in its developmental process. Research shows work-oriented philosophy in Western classical music (Zhang & Li, 2025). This philosophy places performers in secondary positions (Ford, 2020). It forms implicit mechanisms of cultural assimilation (Ford, 2020). In contrast, the Chinese traditional music education system has experienced a transformation from traditional oral transmission to a systematic and standardized teaching system. The introduction of Western music theory occurred in the early 20th century (Tan & Sin, 2021). Western instruments profoundly influenced Chinese traditional music teaching methods (Tan & Sin, 2021). Teaching content has been significantly transformed by this Western influence (Tan & Sin, 2021). Meng's comparative study (2022) analyzes the Orff teaching method as a major global system. The study finds significant advantages in stimulating children's imagination (Meng, 2022). China's new system demonstrates unique value in cultural integration (Meng, 2022).

Contemporary cross-cultural studies reveal teaching method differences (Bautista et al., 2021). Evaluation systems vary significantly across education models (Bautista et al., 2021). Cultural adaptability shows marked differences between systems (Bautista et al., 2021). Comparative analysis of Hong Kong and Spain preschool music education provides insights. Cultural background profoundly influences activity type selection (Bautista et al., 2021). Implementation methods differ based on regional characteristics (Bautista et al., 2021). Although the globalization process has promoted the homogenization trend of preschool education curriculum frameworks, regional cultural characteristics still play a key role in music education practice. Technological development brings revolutionary changes to music education (Ma & Wang, 2023). Virtual reality redefines music teaching experiences (Ma & Wang, 2023). Artificial intelligence promotes personalized learning pathways (Ma & Wang, 2023). However, cross-cultural music cognition requires broader perspectives (Jacoby et al., 2020). Research must go beyond Western-centric approaches (Jacoby et al., 2020). Interdisciplinary collaboration helps understand unique perception mechanisms

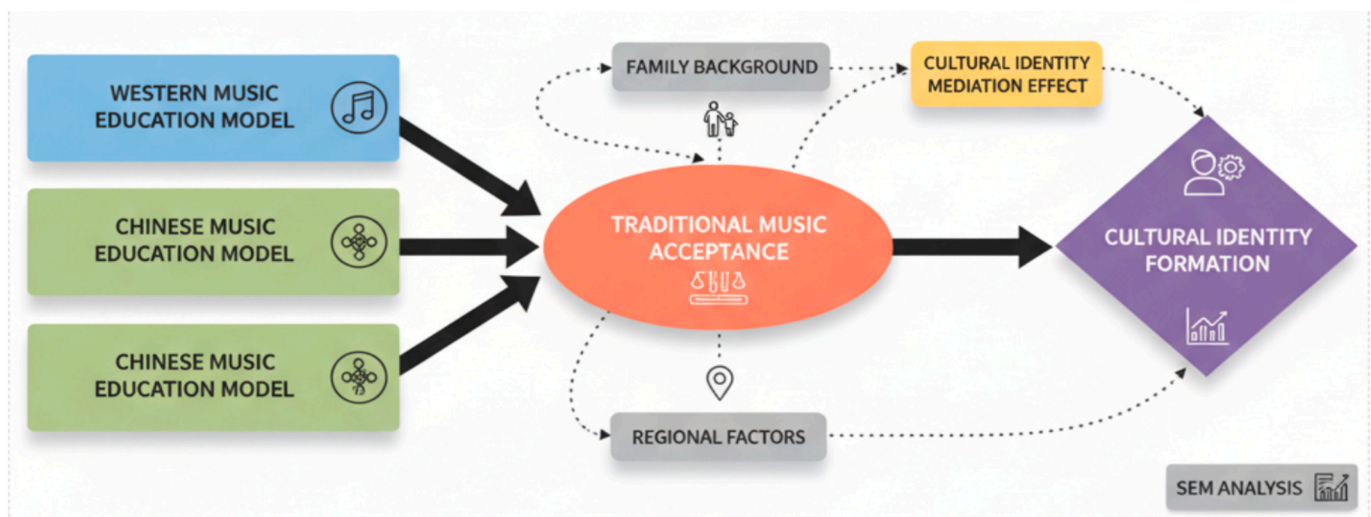


Fig. 1. Research Framework for Traditional Music Acceptance and Cultural Identity under Chinese and Western Music Education Models.

(Jacoby et al., 2020).

2.2. The relationship between cultural identity theory and music education

Cultural identity, as a core element of individual and group identity construction, plays a crucial role in the music education process. Music is not only an important carrier of cultural expression but also a key mechanism for cultural inheritance and identity formation (Crooke et al., 2024). Research from the perspective of social identity theory shows that collective identity is an important component of self-concept, and the relationship between collective self-esteem and individual well-being shows significant differences across different cultural backgrounds. This cultural specificity has important significance in music education practice (Verkuyten, 2005). Yu and Chuangprakhon's (2025) research emphasizes that integrating traditional music into school curricula can promote the cultivation of cultural awareness, strengthen ethnic identity, and ensure the intergenerational transmission of cultural heritage. Music, as a cultural inheritance system, provides survival advantages at the group level through the transmission mechanisms of symbols, meanings, and values, promoting the formation of social cohesion (Rehfeldt et al., 2021).

Theoretical framework research on cross-cultural music participation further reveals the unique role of music in promoting understanding and integration between different cultural groups. Guan et al. (2023), through empirical research on adolescents in multi-ethnic regions of China, found that culturally responsive music teaching can effectively cultivate students' ethnic identity while enhancing their positive influence on Chinese national identity and intergroup attitudes. However, critical perspective research points out that cultural competence approaches have limitations in addressing core issues of cross-cultural tensions, requiring more critical examination of structural factors such as whiteness, colonialism, and white supremacy (Young, 2016). Teachers' cultural awareness and teaching strategies in music education play a key role in students' cultural identity formation, and the implementation of inclusive pedagogy needs to consider the interaction of school-specific policies, structures, culture, and values (Burnard et al., 2008).

2.3. Factors influencing the acceptance of Chinese traditional music

The acceptance of Chinese traditional music in contemporary educational environments is influenced by multiple complex factors, with globalization and the development of modern entertainment industries posing major challenges to the preservation and transmission of traditional music (Norton, 2010). Yang and Welch's (2023) systematic literature review of Chinese music education research from 2007 to 2019 shows that education reform, aesthetic education, Chinese traditional music, and cultural identity are the most prominent themes in formal music education at all levels. Research found that government financial support, policy priorities, curriculum implementation, and teaching innovation are widely considered key measures to address the unsatisfactory current state of music education (Yang & Welch, 2023). Chan's (2018) research emphasizes that achieving sustainable development of traditional music cultural heritage through innovative means requires balancing cultural authenticity with the integration of modern teaching technologies.

The quality of educational resources and teacher training directly affects the teaching effectiveness of traditional music. Research has found that customized educational resources, systematic teacher training, and supportive policies are key elements in preserving and transmitting traditional music (Crawford, 2020). Family background plays an important moderating role in traditional music acceptance, with parents' musical background, family music environment, and early music exposure experiences significantly influencing students' attitudes and interest in traditional music (Creech & Hallam, 2011). Additionally,

research on the integration of music and movement education shows that multimodal teaching methods can enhance music understanding and promote the development of creativity, social well-being, and cultural inclusivity (Del Barrio & Arús, 2024). The application of digital technology provides new possibilities for traditional music education, with innovative forms such as online competitions, virtual environments, and multimedia platforms effectively promoting the sharing and dissemination of multicultural music (Karkina et al., 2022).

2.4. Theoretical framework construction and research hypotheses

Based on comprehensive literature analysis, this study constructs an integrated theoretical framework. The framework examines how education models influence traditional music acceptance. It also explores the role of cultural identity as a key mediator. This framework integrates core elements such as education model characteristics, cultural identity dimensions, traditional music acceptance, and moderating variables (as shown in Fig. 2). The theoretical model builds on established theories. Social identity theory provides the foundational perspective (Tajfel & Turner, 1979). Cultural inheritance theory explains transmission mechanisms (Berry, 2005). Broaden-and-build theory illuminates psychological pathways (Fredrickson, 2001). These theories reveal complex relationship mechanisms between variables (Wang, 2023). The study employs mediation analysis to explore cultural identity's role. This pathway connects education models and traditional music acceptance (Baron & Kenny, 1986). The analytical method has special value in educational research (Ballen & Salehi, 2021). Learning outcomes often stem from complex multi-variable relationships (Ballen & Salehi, 2021).

The selection of mediation analysis as the primary analytical approach is justified by several theoretical and methodological considerations. First, the theoretical framework explicitly posits cultural identity as an intervening mechanism through which education models exert their influence on traditional music acceptance, making mediation analysis the most appropriate method for testing this hypothesized causal chain (Hayes, 2018). Second, mediation analysis allows for the quantification of indirect effects, which is essential for understanding how education models operate through cultural shaping processes rather than solely through direct knowledge transmission. This decomposition of total effects into direct and indirect components provides nuanced insights into the relative importance of different influence pathways. Third, recent methodological advances in bootstrap-based inference provide robust testing of indirect effects even when the assumption of multivariate normality is violated, which is particularly important given the complexity of psychological constructs like cultural identity (Preacher & Hayes, 2008). Fourth, the moderated mediation framework extends basic mediation by allowing examination of whether the indirect effect varies across different conditions (e.g., family background, regional context), which aligns with our theoretical expectation that contextual factors shape the cultural identity formation process. Alternative analytical approaches such as simple regression or basic path analysis would not adequately capture the hierarchical and conditional nature of our theoretical model, as they cannot simultaneously estimate multiple mediating pathways and their interactions with moderating variables.

Based on theoretical analysis and empirical research needs, this study proposes the following research hypotheses:

H1. Western music education models and Chinese traditional music education models show significant differences in students' traditional music acceptance, with students under Chinese traditional music education models demonstrating higher traditional music acceptance (Johnson & Liu, 2022).

H2. Cultural identity plays a mediating role between music education models and traditional music acceptance, meaning education models influence students' acceptance of traditional music by affecting their cultural identity (Yang et al., 2022).

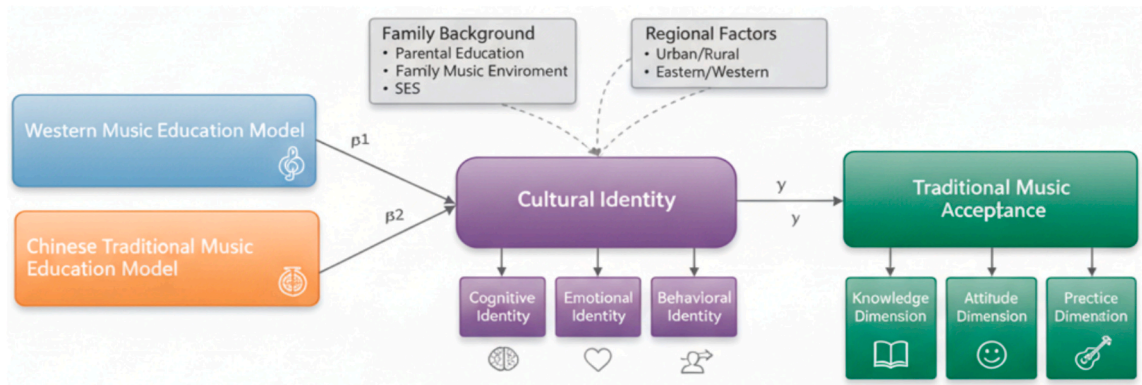


Fig. 2. Comprehensive Theoretical Framework for How Chinese and Western Music Education Models Influence Traditional Music Acceptance and Cultural Identity.

- H3.** Family background (including parental education level, family music environment, and socioeconomic status) moderates the relationship between music education models and cultural identity, with students having rich family music resources showing stronger cultural identity under different education models (Wang et al., 2021).
- H4.** Regional factors (urban vs. rural, eastern vs. western) moderate the relationship between cultural identity and traditional music acceptance, with students from different regional backgrounds showing differences in the process of transforming cultural identity into traditional music acceptance (Prest, 2020).
- H5.** Music participation level (including music learning duration, performance experience, and music club participation) positively moderates the mediating effect of cultural identity, with high music participation strengthening the mediating role of cultural identity between education models and traditional music acceptance (Cores-Bilbao et al., 2019).
- H6.** Self-efficacy and self-esteem play a chain mediating role between music education and students' psychological well-being, further influencing their emotional attitudes and behavioral intentions toward

traditional music (Zelenak, 2015).

The theoretical model clearly demonstrates these hypothetical relationships (as shown in Fig. 3). The multi-level framework comprehensively reveals influence mechanisms. It examines how music education models affect traditional music acceptance. The research framework not only focuses on direct effects but also deeply explores the interaction between mediation and moderation effects, providing systematic theoretical support for understanding cultural inheritance and identity formation (María-de-la-O et al., 2021). The comprehensive framework provides theoretical foundations for empirical research and offers important references for music education policy formulation. The framework guides practical reform in music education contexts (Hao et al., 2025). Research has found that rich cross-cultural music participation experiences can significantly enhance cultural understanding abilities, especially when learners are exposed to musical forms beyond their life experiences (Li et al., 2022).

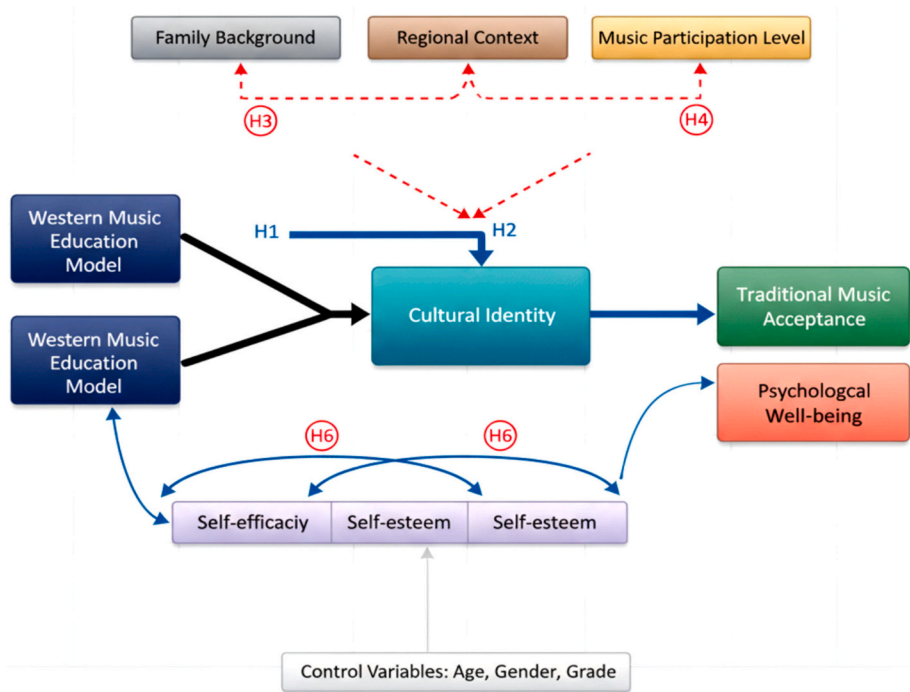


Fig. 3. Theoretical Model of Hypothetical Relationships for How Music Education Models Influence Traditional Music Acceptance.

3. Research design and methods

3.1. Research design and analytical framework

This study adopts mixed research methods, combining qualitative analysis with quantitative research to construct a multi-level empirical research framework for deeply exploring the influencing mechanisms of Chinese and Western music education models on traditional music acceptance and cultural identity. The research design follows the logical path of “theory construction - model specification - empirical testing,” systematically examining the complex relationship network between education models, cultural identity, and traditional music acceptance through cross-sectional data collection and analysis. The analytical framework is built on the foundation of social identity theory and cultural inheritance theory, viewing music education as an important mechanism for cultural transmission. By constructing a comprehensive model containing direct effects, mediating effects, and moderating effects, it reveals the differentiated pathways of student cultural identity formation and traditional music acceptance under different education models (Fig. 4). This framework takes education model characteristics as antecedent variables, cultural identity as a mediating variable, traditional music acceptance as an outcome variable, and incorporates moderating variables such as family background, regional factors, and music participation level, forming a multi-dimensional, multi-level analytical system.

The selection of this mixed research design and multi-level analytical framework is justified by several theoretical and practical considerations. First, the complexity of cultural identity formation necessitates both quantitative measurement of relationships and qualitative understanding of underlying mechanisms; purely quantitative approaches may miss the nuanced ways in which students experience and internalize cultural values through music education. Second, structural equation modeling (SEM) is particularly suited for this research because it simultaneously estimates multiple latent constructs and their interrelationships, which is essential when dealing with abstract psychological concepts like cultural identity that cannot be directly observed

but must be inferred from multiple indicators (Kline, 2016). Unlike traditional regression approaches that analyze relationships one at a time, SEM provides a holistic test of the entire theoretical model while accounting for measurement error, thereby yielding more accurate parameter estimates. Third, the moderated mediation framework is theoretically necessary given our hypotheses about how contextual factors (family background, regional culture, music participation) alter the mediation process; this analytical approach allows testing of conditional indirect effects, revealing for whom and under what circumstances the mediation operates most strongly (Hayes, 2015). Fourth, alternative approaches such as simple regression or basic path analysis would not adequately capture the hierarchical and conditional nature of our theoretical model, as they cannot simultaneously estimate multiple mediating pathways, account for measurement error in latent variables, or test interaction effects between mediators and moderators. Finally, the multi-group comparison capability of SEM enables rigorous testing of whether the hypothesized relationships operate equivalently across different subpopulations (e.g., regions, education model types), which is crucial for understanding the generalizability of findings and developing context-sensitive educational policies.

3.2. Sample selection and data collection

Sample selection employed stratified random sampling method, covering three major geographical regions (eastern, central, and western), and encompassing three city levels (provincial capitals, prefecture-level cities, and county-level cities) to ensure sample representativeness and diversity. The target group was positioned as students who had received systematic music education at junior high school, senior high school, and university levels, with an age range of 12–22 years. Students in this age group are in a critical period of cultural identity formation and have typical perceptions and reactions to music education models.

The stratified random sampling was implemented through a rigorous four-stage procedure to ensure systematic coverage and minimize selection bias. In the first stage of primary stratification by geographic region, schools were categorized into eastern, central, and western

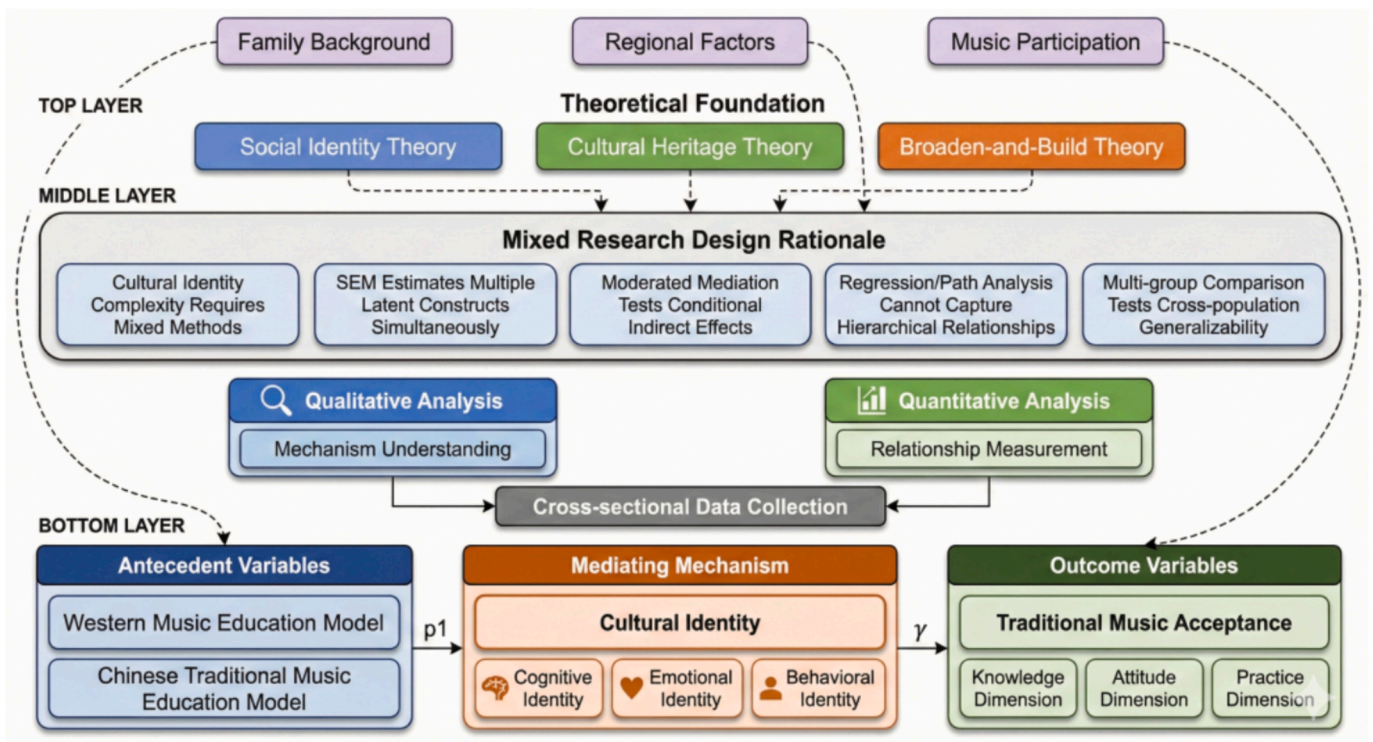


Fig. 4. Research Design and Analytical Framework.

regions based on national administrative classifications issued by the National Bureau of Statistics of China. Within each region, provincial education departments provided comprehensive lists of schools with established music education programs, including both general education schools with music curricula and specialized arts schools. In the second stage of secondary stratification by city level, within each region, schools were further stratified by city tier (provincial capitals, prefecture-level cities, and county-level cities) to ensure adequate representation across the urban-rural continuum and different levels of educational resource availability. This stratification was important because music education resources and traditional music exposure opportunities vary substantially across city tiers. In the third stage of school selection, from each stratum defined by the intersection of region and city level, schools were randomly selected using computer-generated random numbers, with selection probability proportional to enrollment size (PPS sampling). This approach ensured that larger schools with more diverse student populations had appropriately higher selection probabilities while maintaining randomness. A total of 48 schools were selected across the nine strata (3 regions $\times$ 3 city levels), with 4–6 schools per stratum depending on population density. In the fourth stage of student selection, within selected schools, classes with music education courses were first identified through coordination with school administrators. From eligible classes, students were randomly selected using class rosters, with random numbers generated separately for each class to ensure independence of selection.

Inclusion criteria were established as follows: (1) aged between 12 and 22 years; (2) received at least 3 years of systematic music education (either Western, traditional Chinese, or comprehensive); (3) no hearing impairments that would affect music perception; (4) ability to complete questionnaires independently in Chinese; (5) voluntary participation with informed consent (parental consent for participants under 18).

Exclusion criteria included: (1) completion time less than 10 min (indicating insufficient attention); (2) patterned responses detected through attention check items; (3) missing values exceeding 10% of total items; (4) logical contradictions in responses (e.g., claiming no music education experience while reporting extensive music practice); (5) duplicate submissions from the same IP address.

Sample size determination was based on the statistical requirements of structural equation modeling, using the following formula to calculate the minimum sample size based on the quantitative relationship between latent variables and observed variables in the model:

$$n = \max(10 \times q, 200) \quad (1)$$

where n represents the minimum sample size and q represents the number of free parameters to be estimated in the model. Considering the complexity of this research model, which contains 6 latent variables and 32 observed variables with approximately 68 free parameters, the minimum sample size should be no less than 680. To improve statistical power and account for the impact of invalid questionnaires, the study planned to collect 1200 valid questionnaires, with quota allocation by education model type (Western music education model-dominant, Chinese traditional music education model-dominant, comprehensive type), 400 for each type.

To address potential regional and educational-model biases, several control strategies were implemented throughout the sampling and data collection process. For regional bias control, quota monitoring was conducted in real-time during data collection to ensure proportional representation across regions, with a target of 33% ($\pm 5\%$) for each region; any emerging imbalances triggered additional sampling efforts in underrepresented regions. Post-stratification weights were calculated after data collection to adjust for remaining imbalances between sample distributions and known population parameters from national education statistics, ensuring that descriptive statistics and model estimates reflected the target population rather than sampling artifacts. Multi-group measurement invariance testing (detailed results reported in Section

4.6) verified that measurement models functioned equivalently across regions, confirming that observed regional differences reflected true substantive differences rather than measurement artifacts. For education model classification bias control, recognizing that student self-reports of their education model might be subject to misclassification, we implemented a validation procedure in which classifications were verified through examination of official school curriculum documents and semi-structured interviews with music teachers for a randomly selected 15% subsample ($n = 174$). This verification achieved 91.3% agreement with student self-reports (Cohen's $\kappa = 0.84$), indicating substantial reliability of the classification scheme. Cases of disagreement were resolved through researcher review and, where necessary, reclassification based on objective curriculum evidence. Additionally, to minimize social desirability bias in responses, the survey employed neutral framing of questions, assured confidentiality, and included reverse-scored items to detect acquiescence response patterns.

Data collection adopted a combination of online and offline methods, coordinating through education departments to conduct centralized testing in selected schools, ensuring the standardization and validity of data collection. Survey time was controlled at 30–40 min, with professional researchers providing on-site guidance to improve data quality (as shown in Fig. 5).

3.3. Variable measurement and questionnaire design

The measurement of core variables in the study was based on mature scales adapted to the Chinese context to ensure the reliability and validity of measurement instruments. Music education models were operationalized through a classification system that underwent expert validation. Prior to the main study, a panel of 5 music education experts (3 professors and 2 senior practitioners) reviewed the classification criteria. Additionally, a pilot study with 120 students was conducted to validate the classification scheme, achieving an inter-rater reliability of $\kappa = 0.87$.

Music education models were coded using the following calculation formula:

$$MEM_i = \frac{\sum_{j=1}^n w_j \times t_{ij}}{\sum_{j=1}^n t_{ij}} \quad (2)$$

where MEM_i represents the music education model score for student i , w_j represents the weight of music education content type j (Western music = 1, Chinese traditional music = 0, integrated type = 0.5), and t_{ij} represents the time investment of student i in music education type j . The classification basis was established through: (1) curriculum content analysis (>60% Western content = Western-dominant; >60% traditional content = Traditional-dominant; 40–60% of either = Comprehensive); (2) primary instrument type (Western vs. Chinese traditional); (3) theoretical framework studied (Western music theory vs. Chinese music theory).

The education model classification system underwent comprehensive validation to ensure its psychometric soundness. Content validity was established through a two-round Delphi process involving 8 music education experts (4 professors from conservatories specializing in music education theory, and 4 senior practitioners with over 15 years of teaching experience in both Western and traditional Chinese music). In the first round, experts independently evaluated the relevance and clarity of classification criteria using a 4-point scale; items receiving less than 75% agreement were revised or eliminated. In the second round, revised criteria achieved consensus with greater than 80% agreement on all classification dimensions. Criterion validity was assessed by correlating MEM scores calculated from student self-reports with objective curriculum records obtained from 12 pilot schools that provided detailed documentation of their music education programs. The

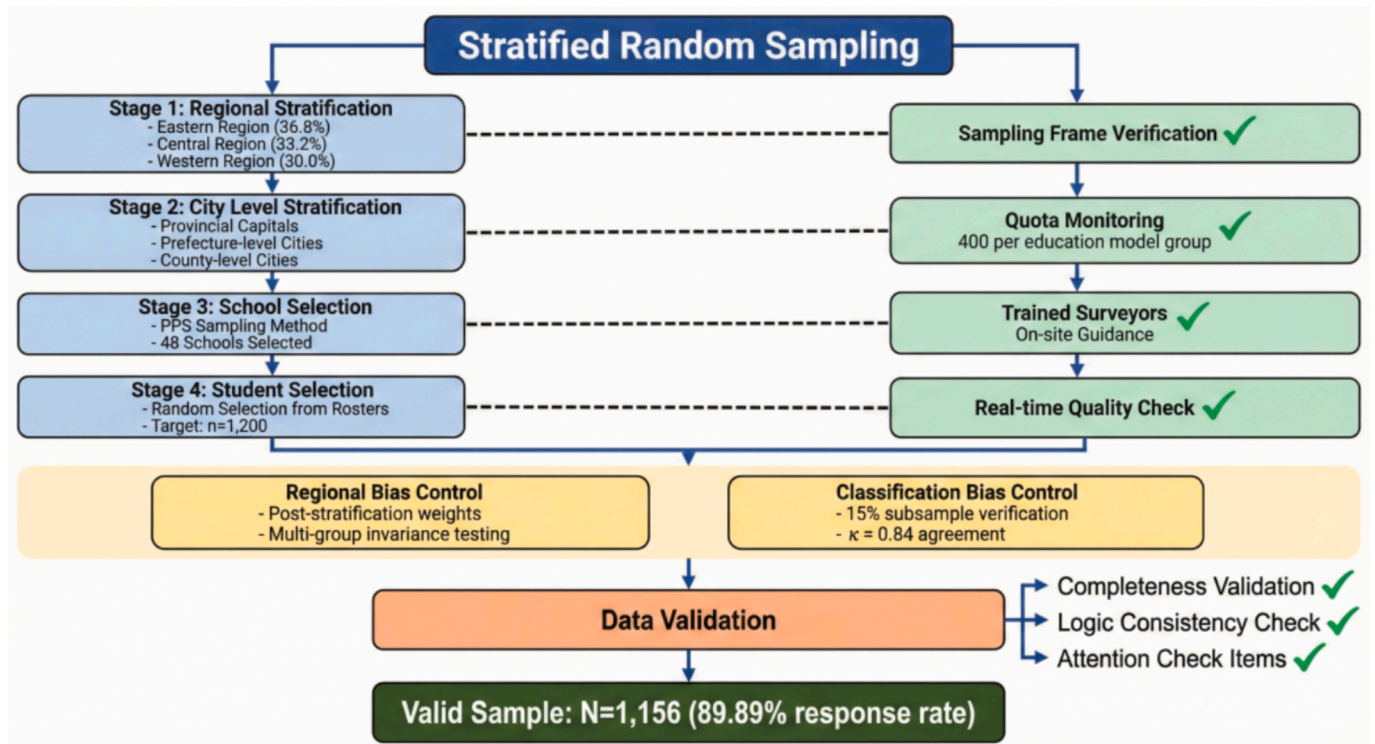


Fig. 5. Data Collection Process and Quality Control System.

correlation between self-reported MEM scores and curriculum-based classifications was $r = 0.84$ ($p < 0.001$), indicating strong criterion validity. Test-retest reliability was evaluated by administering the classification questionnaire twice to a subsample of 86 students over a 4-week interval, yielding an intraclass correlation coefficient (ICC) of 0.89 (95% CI [0.82, 0.94]), demonstrating excellent temporal stability. Construct validity was further supported by the finding that MEM scores significantly differentiated students attending specialized conservatories (predominantly traditional or Western-focused) from those in general education schools with comprehensive music programs, $F(2, 117) = 47.23$, $p < 0.001$, $\eta^2 = 0.45$.

Cultural identity measurement adopted a multi-dimensional scale including cognitive identity, emotional identity, and behavioral identity dimensions, with 5–7 items for each dimension, measured using a 7-point Likert scale. The dimension weights (α , β , γ) were determined through exploratory factor analysis (EFA) using principal component analysis with varimax rotation on a calibration sample ($n = 580$). The EFA revealed a three-factor structure explaining 67.4% of total variance. Confirmatory factor analysis (CFA) on a validation sample ($n = 576$) confirmed the factor structure ($\chi^2/df = 2.13$, CFI = 0.95, TLI = 0.94, RMSEA = 0.044). The comprehensive cultural identity score was calculated using the following formula:

$$CI_i = \alpha_1 \cdot COG_i + \alpha_2 \cdot EMO_i + \alpha_3 \cdot BEH_i \quad (3)$$

where CI_i represents the total cultural identity score for student i , COG_i , EMO_i , and BEH_i represent cognitive, emotional, and behavioral identity scores respectively, and $\alpha_1 = 0.35$, $\alpha_2 = 0.38$, $\alpha_3 = 0.27$ are dimension weight coefficients determined through factor analysis, satisfying $\alpha_1 + \alpha_2 + \alpha_3 = 1$.

The cultural identity scale was adapted from established instruments in the cross-cultural psychology literature (Phinney, 1992; Verkuyten, 2005) with systematic modifications to enhance relevance to the music education context. The adaptation process began with item development involving focus group discussions with 36 students representing all three education model types (12 from each type), who provided feedback on

item clarity, cultural appropriateness, and relevance to their music education experiences. Based on focus group insights, initial items were drafted to capture how students' musical experiences shaped their sense of Chinese cultural belonging. An initial pool of 28 items was subjected to rigorous item analysis on a pilot sample ($n = 156$). Items with corrected item-total correlations below 0.40 were eliminated, and items showing ceiling or floor effects (more than 80% of responses at extreme points) were revised or removed, resulting in a refined 18-item scale. The dimension weights ($\alpha_1 = 0.35$ for cognitive identity, $\alpha_2 = 0.38$ for emotional identity, $\alpha_3 = 0.27$ for behavioral identity) were derived from EFA on a calibration sample ($n = 580$) using principal component analysis with varimax rotation; these weights reflected the relative contribution of each dimension to the overall cultural identity construct as indicated by eigenvalues and factor loadings. The derived weights were then cross-validated on a holdout sample ($n = 576$) by comparing model fit between the weighted composite approach and an unweighted approach, with the weighted approach showing superior fit ($\Delta CFI = 0.012$, $\Delta RMSEA = -0.006$). Critically, measurement invariance across education model groups was rigorously tested following established protocols (Chen, 2007). Configural invariance (same factor structure across groups) was supported ($\Delta CFI = 0.002$ compared to baseline). Metric invariance (equal factor loadings) was confirmed ($\Delta CFI = 0.005$, below the 0.01 threshold). Scalar invariance (equal item intercepts) was also supported ($\Delta CFI = 0.008$). These results indicate that the cultural identity construct was measured equivalently across students receiving different types of music education, allowing meaningful comparison of latent means and structural relationships across groups. The validation process and results are summarized in Table 1.

Traditional music acceptance measurement covered three levels: knowledge mastery, attitude tendency, and practical participation, evaluated through a comprehensive indicator system. Similarly, the weights for these dimensions (β_1 , β_2 , β_3) were established through EFA ($\beta_1 = 0.32$ for knowledge, $\beta_2 = 0.36$ for attitude, $\beta_3 = 0.32$ for practice), with the factor structure validated through CFA.

$$TMA_i = \beta_1 \cdot KNO_i + \beta_2 \cdot ATT_i + \beta_3 \cdot PRA_i \quad (4)$$

Table 1
Measurement Validation Results for Cultural Identity Scale.

Validation Aspect	Method	Result	Criterion	Conclusion
Content Validity	Delphi Process (8 experts, 2 rounds)	>80% agreement	>75%	Supported
Item Analysis	Corrected item-total correlation	Range: 0.42–0.71	>0.40	All items retained
Factor Structure (EFA)	Principal component, varimax	3 factors, 67.4% variance	>60%	Supported
Factor Structure (CFA)	Maximum likelihood	$\chi^2/df = 2.13$, CFI = 0.95, RMSEA = 0.044	CFI > 0.90, RMSEA < 0.08	Supported
Internal Consistency	Cronbach's α	Cognitive = 0.84, Emotional = 0.87, Behavioral = 0.81, Total = 0.89	>0.70	Excellent
Composite Reliability	CR	Cognitive = 0.85, Emotional = 0.88, Behavioral = 0.82	>0.70	Excellent
Convergent Validity	AVE	Cognitive = 0.58, Emotional = 0.64, Behavioral = 0.53	>0.50	Supported
Discriminant Validity	Fornell-Larcker criterion	$\sqrt{AVE} > \text{inter-factor correlations}$	$\sqrt{AVE} > r$	Supported
Configural Invariance	Multi-group CFA	$\Delta CFI = 0.002$	<0.01	Supported
Metric Invariance	Multi-group CFA	$\Delta CFI = 0.005$	<0.01	Supported
Scalar Invariance	Multi-group CFA	$\Delta CFI = 0.008$	<0.01	Supported

where TMA_i represents traditional music acceptance, KNO_i , ATT_i , and PRA_i are knowledge, attitude, and practice dimension scores respectively, and β_1 , β_2 , β_3 are dimension weight coefficients.

3.4. Empirical model construction

The structural equation model constructed in the study includes two parts: measurement model and structural model. The measurement model describes the relationship between latent variables and observed variables, while the structural model depicts the causal paths between latent variables. The basic measurement equations are expressed as:

$$X = \Lambda_x \xi + \delta \quad (5)$$

$$Y = \Lambda_y \eta + \varepsilon \quad (6)$$

where X is the vector of exogenous observed variables, Y is the vector of endogenous observed variables, ξ represents exogenous latent variables, η represents endogenous latent variables, Λ_x and Λ_y are factor loading matrices, and δ and ε are measurement errors.

The basic equation of the structural model is:

$$\eta = B\eta + \Gamma\xi + \zeta \quad (7)$$

where B is the path coefficient matrix between endogenous latent variables, Γ is the effect coefficient matrix of exogenous latent variables on

endogenous latent variables, and ζ is the residual vector.

Mediation effect testing employed the Bootstrap method, judging the significance of mediation through calculating confidence intervals for indirect effects. We used 5000 bootstrap samples with bias-corrected and accelerated (BCa) confidence intervals. The calculation formula for indirect effects is:

$$IE = a \times b \quad (8)$$

where a represents the effect of the independent variable on the mediating variable, b represents the effect of the mediating variable on the dependent variable, and the significance of indirect effect IE is judged by whether the 95% confidence interval contains 0. Additionally, we calculated the effect size measure κ^2 for the indirect effect following Preacher and Kelley (2011) guidelines.

Prior to SEM estimation, comprehensive assumption testing was conducted to ensure the validity of statistical inferences. Univariate normality was assessed for all observed variables using skewness and kurtosis statistics. All skewness values fell within the range of -1.23 to 1.47 (all $|\text{values}| < 2$), and all kurtosis values ranged from -0.89 to 2.34 (all $|\text{values}| < 7$), meeting the recommended thresholds for acceptable departure from normality in SEM applications (Kline, 2016). Multivariate normality was evaluated using Mardia's coefficient of multivariate kurtosis, which yielded a value of 36.42. Although this slightly exceeded the critical value for perfect multivariate normality (critical value ≈ 33 for our number of variables), the magnitude of departure was modest. To address this minor violation, we employed robust maximum likelihood estimation (MLR) with Satorra-Bentler scaled chi-square corrections, which provides unbiased parameter estimates and accurate standard errors even under conditions of non-normality (Satorra & Bentler, 2001). Multicollinearity among predictor variables was examined through multiple diagnostics. Variance inflation factors (VIF) were calculated for all predictors in preliminary regression models, with values ranging from 1.23 to 2.87, all well below the conventional threshold of 5 (or the more conservative threshold of 10), indicating no serious multicollinearity concerns. Condition indices were also examined, with the highest value being 18.4, below the critical threshold of 30 that would indicate problematic collinearity. Bivariate correlations among latent variables ranged from 0.21 to 0.58, none approaching the level (>0.85) that would suggest construct redundancy. The complete assumption testing results are presented in Table 2.

Measurement invariance across the three education model groups was established through sequential testing following Chen's (2007) recommended protocol, which is essential for ensuring that observed group differences in structural paths reflect true substantive differences rather than measurement artifacts. Configural invariance, testing

Table 2
Assumption Testing Results for Structural Equation Modeling.

Assumption	Test Method	Result	Criterion	Conclusion
Univariate Normality - Skewness	Descriptive statistics	Range: $[-1.23, 1.47]$	$ \text{skew} < 2$	Satisfied
Univariate Normality - Kurtosis	Descriptive statistics	Range: $[-0.89, 2.34]$	$ \text{kurt} < 7$	Satisfied
Multivariate Normality	Mardia's coefficient	36.42	Critical ≈ 33	Minor violation; MLR used
Multicollinearity - VIF	Regression diagnostics	Range: $[1.23, 2.87]$	$VIF < 5$	No issue
Multicollinearity - Condition Index	Regression diagnostics	Maximum: 18.4	$CI < 30$	No issue
Linearity	Residual plots	No curvilinear patterns	Visual inspection	Satisfied
Homoscedasticity	Breusch-Pagan test	$\chi^2(6) = 8.73$, $p = 0.189$	$p > 0.05$	Satisfied

whether the same factor structure holds across groups, was first established ($\chi^2 = 486.3$, $df = 144$, $CFI = 0.952$, $RMSEA = 0.048$). Metric invariance, constraining factor loadings to be equal across groups, was then tested; the change in fit was minimal ($\Delta\chi^2 = 12.4$, $\Delta df = 8$, $p = 0.134$, $\Delta CFI = 0.003$), supporting the equivalence of factor loadings. Scalar invariance, additionally constraining item intercepts, showed a slight but acceptable decrease in fit ($\Delta\chi^2 = 22.7$, $\Delta df = 8$, $p = 0.012$, $\Delta CFI = 0.009$). Following Cheung and Rensvold's (2002) recommendation that ΔCFI values below 0.01 indicate invariance, all levels of measurement invariance were supported, confirming that the constructs were measured equivalently across education model groups and allowing meaningful comparison of latent means and structural paths. The invariance testing results are summarized in Table 3.

Moderation effects were tested by constructing interaction terms, with the moderation model expressed as:

$$Y = \beta_0 + \beta_1 X + \beta_2 M + \beta_3 (X \times M) + \varepsilon \quad (9)$$

where X is the independent variable, M is the moderating variable, $X \times M$ is the interaction term, and the significance of β_3 indicates whether a moderation effect exists.

Multi-group analysis was used to test model equivalence across different groups, comparing nested models by constraining parameters across groups, with the chi-square difference test formula:

$$\Delta\chi^2 = \chi^2_{\text{constrained}} - \chi^2_{\text{unconstrained}} \quad (10)$$

When $\Delta\chi^2$ is significant, it indicates significant differences between groups (as shown in Fig. 6).

3.5. Data analysis methods

Data analysis adopted a strategy combining multiple statistical methods, using SPSS 26.0 for descriptive statistics, correlation analysis, and preliminary regression analysis, Mplus 8.3 for structural equation model analysis and multi-group comparison, and Process macro for testing mediation and moderation effects.

Reliability testing employed Cronbach's α coefficient and composite reliability (CR) for evaluation, while validity testing included comprehensive evaluation of content validity, construct validity, and discriminant validity. Model fit evaluation used multiple indicators including chi-square to degrees of freedom ratio (χ^2/df), comparative fit index (CFI), Tucker-Lewis index (TLI), root mean square error of approximation (RMSEA), and standardized root mean square residual (SRMR), with fit criteria set as: $\chi^2/df < 3$, $CFI > 0.90$, $TLI > 0.90$, $RMSEA < 0.08$, $SRMR < 0.08$.

The selection of these specific fit criteria and the use of multiple fit indices follows current methodological best practices in structural equation modeling (Hu & Bentler, 1999; Kline, 2016; Marsh et al., 2004). The chi-square to degrees of freedom ratio (χ^2/df) is reported because the chi-square test itself is highly sensitive to sample size, often rejecting well-fitting models in large samples like ours; the ratio provides a sample-size-adjusted assessment with values below 3 generally indicating acceptable fit. CFI and TLI are incremental fit indices that compare the target model against a baseline (null) model; they are relatively insensitive to sample size and provide assessment of how much better the hypothesized model fits compared to a model assuming no relationships among variables. The choice of 0.90 as the threshold follows traditional conventions, though values above 0.95 indicate

excellent fit. RMSEA is an absolute fit index that evaluates model fit per degree of freedom, incorporating a parsimony correction that penalizes model complexity; the criterion of 0.08 represents adequate fit, while values below 0.05 indicate close fit, and the 90% confidence interval provides an uncertainty estimate. SRMR represents the average discrepancy between observed and model-implied correlations, with lower values indicating better fit; the 0.08 threshold is well-established. Using multiple indices from different families (absolute, incremental, parsimony-adjusted) reduces the risk of model acceptance or rejection based on a single potentially misleading indicator, as different indices are sensitive to different types of model misspecification. This multi-index approach ensures robust evaluation of model fit from complementary perspectives.

Common method bias control combined procedural and statistical controls. Procedural controls included anonymous responses, reverse-scored items, and randomized item order. Statistical control employed Harman's single-factor test, with the first factor's variance explanation rate required to be below 40%.

Missing value treatment employed multiple imputation method, analyzing by creating multiple complete datasets and then integrating analysis results to improve parameter estimation accuracy. The imputation model included all analysis variables and auxiliary variables related to the missing mechanism, with 20 imputations to ensure estimation stability.

Model robustness testing was achieved through multiple methods, including comparative analysis using different estimation methods (maximum likelihood estimation versus robust maximum likelihood estimation), using Bootstrap method (5000 resamples) to calculate standard errors and confidence intervals for parameter estimates, conducting sensitivity analysis to examine the impact of extreme values on results, and evaluating model predictive power through cross-validation. The comprehensive use of these analytical methods ensured the reliability and generalizability of research conclusions (as shown in Fig. 7).

4. Empirical analysis results

4.1. Descriptive statistics and sample characteristic analysis

This study collected a total of 1286 questionnaires. After eliminating invalid questionnaires with patterned responses, missing values exceeding 10%, and logical contradictions, 1156 valid questionnaires were obtained, with an effective response rate of 89.89%. The demographic characteristic distribution of the sample showed good representativeness, with males accounting for 46.3% and females 53.7%, showing relatively balanced gender distribution. Age distribution showed normal characteristics, with 31.2% aged 12–15, 38.7% aged 16–18, and 30.1% aged 19–22, covering the critical period of adolescent cultural identity formation (Table 4).

The sample showed clear differentiation characteristics in music learning experience. Students receiving primarily Western music education accounted for 32.7%, mainly learning Western instruments such as piano and violin, with curriculum content centered on Western music theory, sight-singing and ear training, and Western music history. Students receiving primarily Chinese traditional music education accounted for 35.6%, mainly learning ethnic instruments such as erhu, guzheng, and pipa, with curriculum covering Chinese music history, ethnic music theory, and opera appreciation. Students adopting

Table 3
Measurement Invariance Testing Across Education Model Groups.

Model	χ^2	df	CFI	RMSEA	SRMR	$\Delta\chi^2$	Δdf	p	ΔCFI
Configural	486.3	144	0.952	0.048	0.039	–	–	–	–
Metric	498.7	152	0.949	0.047	0.042	12.4	8	0.134	0.003
Scalar	521.4	160	0.943	0.048	0.045	22.7	8	0.012	0.009

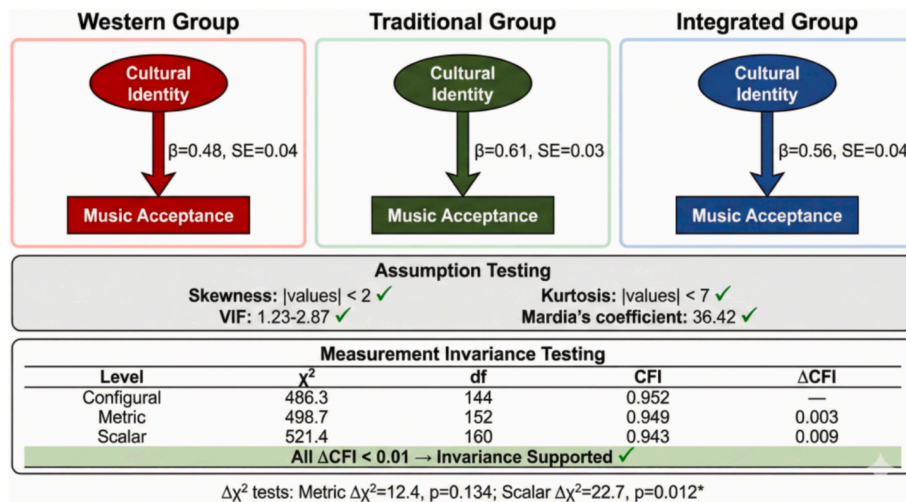


Fig. 6. Structural Equation Model Paths and Multi-group Analysis Framework.

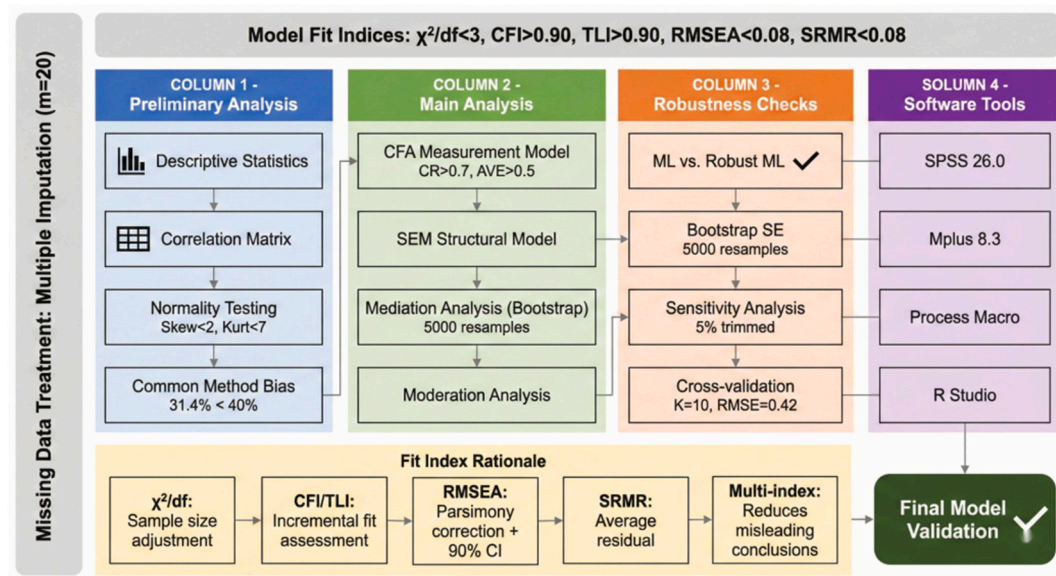


Fig. 7. Data Analysis Method System and Robustness Testing Process.

Table 4
Sample Basic Characteristic Distribution ($N = 1156$).

Variable Category	Classification	Frequency	Percentage (%)	Cumulative Percentage (%)
Gender	Male	535	46.3	46.3
	Female	621	53.7	100
Age Group	12–15 years	361	31.2	31.2
	16–18 years	447	38.7	69.9
	19–22 years	348	30.1	100
Region	Eastern	425	36.8	36.8
	Central	384	33.2	70
	Western	347	30	100
Education Model	Western music-dominant	378	32.7	32.7
	Traditional music-dominant	412	35.6	68.3
	Comprehensive	366	31.7	100

comprehensive education models accounted for 31.7%, simultaneously exposed to both Chinese and Western music systems, forming a relatively balanced music knowledge structure (as shown in Fig. 8).

Descriptive statistical analysis of core variables revealed basic distribution characteristics and correlations between variables. The mean cultural identity score was 4.82 ($SD = 1.13$), and the mean traditional music acceptance was 4.56 ($SD = 1.27$), both showing moderately high levels but with large individual differences (Table 5).

4.2. Differences in students' traditional music acceptance under Chinese and Western music education models

Analysis of differences in students' traditional music acceptance under different education models showed significant group effects (as shown in Table 6). One-way ANOVA results indicated significant differences in traditional music acceptance among students under the three education models ($F(2,1153) = 78.42, p < 0.001, \eta^2 = 0.12$). Post-hoc comparisons using the Bonferroni method for multiple comparison correction showed that students receiving primarily traditional music education ($M = 5.24, SD = 0.98$) had significantly higher traditional music acceptance than comprehensive education model students ($M = 4.58, SD = 1.12, p < 0.001, \text{Cohen's } d = 0.65$), who in turn had significantly higher acceptance than students receiving primarily Western

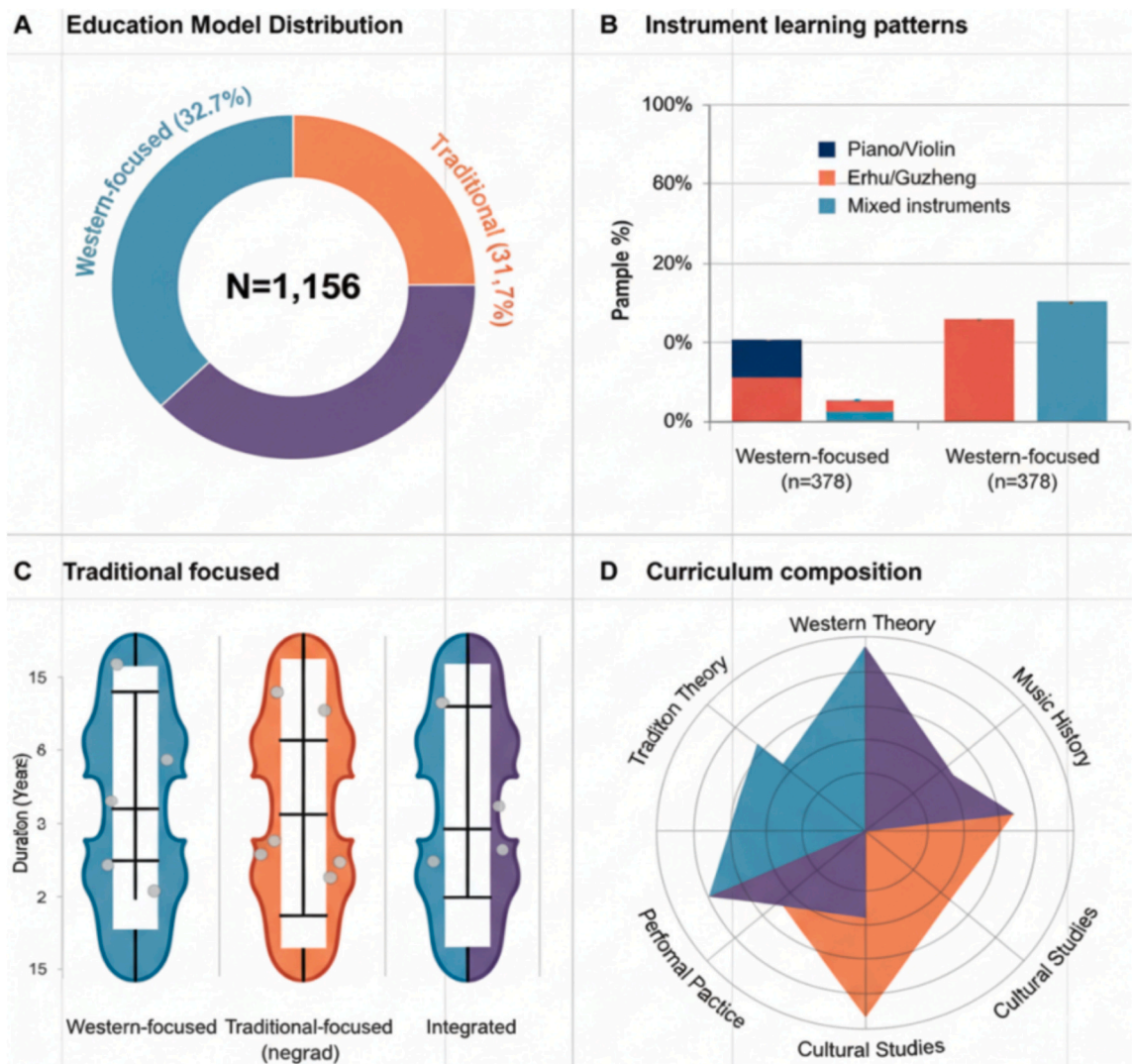


Fig. 8. Sample Music Education Background Distribution Characteristics.

Table 5
Core Variable Descriptive Statistics and Correlation Matrix.

Variable	M	SD	1	2	3	4	5	6
1. Education Model	0.48	0.35	1					
2. Cultural Identity	4.82	1.13	−0.32**	1				
3. Traditional Music Acceptance	4.56	1.27	−0.41**	0.58**	1			
4. Family Background	3.89	1.05	0.28**	0.31**	0.35**	1		
5. Regional Factors	0.51	0.42	−0.15**	0.22**	0.26**	0.18**	1	
6. Music Participation Level	4.23	1.18	0.35**	0.42**	0.48**	0.37**	0.21**	1

Note: $p < 0.01$; Education model coding: 0 = Traditional music-dominant, 0.5 = Comprehensive, 1 = Western music-dominant.

Table 6
ANOVA of Traditional Music Acceptance under Different Education Models.

Source of Variation	SS	df	MS	F	p	η^2	95% CI for η^2
Between groups	234.56	2	117.28	78.42	<0.001	0.12	[0.08, 0.16]
Within groups	1724.38	1153	1.5				
Total	1958.94	1155					

music education ($M = 3.87$, $SD = 1.35$, $p < 0.001$, Cohen's $d = 0.58$).
In-depth analysis of the three dimensions of traditional music acceptance revealed that education models had the most significant impact on the knowledge dimension ($F = 92.31$, $p < 0.001$, $\eta^2 = 0.14$,

95% CI [0.10, 0.18]), followed by the practice dimension ($F = 71.25$, $p < 0.001$, $\eta^2 = 0.11$, 95% CI [0.07, 0.15]), with relatively smaller but still statistically significant differences in the attitude dimension ($F = 45.67$, $p < 0.001$, $\eta^2 = 0.07$, 95% CI [0.04, 0.11]). This differentiated impact

pattern indicates that education models primarily influence students' acceptance of traditional music through knowledge transmission and practical training, while attitude-level changes require deeper cultural identity transformation (as shown in Fig. 9).

4.3. Testing the mediating effect of cultural identity

Structural equation model analysis results supported the mediating role hypothesis of cultural identity between education models and traditional music acceptance, with model fit indicators reaching good levels ($\chi^2(df = 486) = 1089.42$, $p < 0.001$; $\chi^2/df = 2.24$; CFI = 0.945; TLI = 0.937; RMSEA = 0.033 [90% CI: 0.030, 0.036]; SRMR = 0.041). Path analysis showed that education models had a significant negative effect on cultural identity ($\beta = -0.38$, $SE = 0.04$, $p < 0.001$, 95% CI [-0.46, -0.30]), indicating that the higher the proportion of Western music education, the lower students' Chinese cultural identity level. Cultural identity had a significant positive effect on traditional music acceptance ($\beta = 0.52$, $SE = 0.03$, $p < 0.001$, 95% CI [0.46, 0.58]), with students having higher cultural identity levels showing stronger willingness to accept traditional music (as shown in Table 7).

Bootstrap method (5000 resamples) test results showed that the 95% confidence interval for the indirect effect did not contain 0 [-0.247, -0.149]), confirming the partial mediating role of cultural identity, with the mediation effect accounting for 47.7% of the total effect. The effect size $\kappa^2 = 0.126$ (95% CI [0.094, 0.162]) indicates a medium-sized indirect effect. This result indicates that education models not only directly influence traditional music acceptance but also indirectly influence it through shaping students' cultural identity, with cultural identity playing an important bridging role in this process.

Further moderated mediation analysis found that the mediating effect of cultural identity showed significant differences under different conditions. When students had higher levels of music participation, the mediating effect of cultural identity was significantly enhanced (Index of Moderated Mediation = 0.082, $SE = 0.021$, 95% CI [0.041, 0.123]), indicating that active music practice participation can strengthen the transformation process from cultural identity to traditional music acceptance (as shown in Fig. 10).

4.4. Moderating variable analysis: family background and regional factors

The moderating role of family background in the relationship between education models and cultural identity received empirical support. Interaction term testing (Table 8) showed that family background significantly moderated the impact of education models on cultural identity ($\beta = 0.142$, $SE = 0.031$, $p < 0.001$, 95% CI [0.081, 0.203]). Simple slope analysis showed that under conditions of rich family music

Table 7

Mediation Effect Decomposition and Bootstrap Test Results.

Path	Effect Value	SE	95% CI Lower	95% CI Upper	p-value	κ^2
Total Effect	-0.415	0.042	-0.497	-0.333	<0.001	-
Direct Effect	-0.217	0.038	-0.291	-0.143	<0.001	-
Indirect Effect	-0.198	0.025	-0.247	-0.149	<0.001	0.126
Mediation Effect Proportion	47.70%	-	-	-	-	-

resources (+1SD), the negative impact of education models on cultural identity was weakened ($\beta = -0.24$, $p < 0.001$, 95% CI [-0.32, -0.16]), while under conditions of poor family music resources (-1SD), this negative impact was significantly strengthened ($\beta = -0.52$, $p < 0.001$, 95% CI [-0.60, -0.44]).

Regional factor moderation analysis revealed regional differences in cultural identity transformation mechanisms (as shown in Table 9). The effect of cultural identity on traditional music acceptance for eastern region students ($\beta = 0.48$, $p < 0.001$, 95% CI [0.40, 0.56]) was significantly lower than for central region ($\beta = 0.56$, $p < 0.001$, 95% CI [0.48, 0.64]) and western region students ($\beta = 0.61$, $p < 0.001$, 95% CI [0.53, 0.69]). Multi-group analysis constrained model comparison confirmed the statistical significance of these differences ($\Delta\chi^2 = 18.73$, $\Delta df = 2$, $p < 0.001$).

These regional differences reflect the deep influence of different regional cultural environments on music education effects. The relatively strong traditional cultural atmosphere in western regions facilitates the transformation of cultural identity into actual acceptance of traditional music, while eastern regions have higher levels of internationalization, with students having more opportunities to encounter diverse cultures, making the transformation process from cultural identity to acceptance of specific musical forms relatively complex (as shown in Fig. 11).

4.5. Structural equation model validation

The complete structural equation model integrated direct effects, mediating effects, and moderating effects, forming a complex relationship network. The overall model fit indicators performed well ($\chi^2(df = 486) = 1089.42$, $p < 0.001$; $\chi^2/df = 2.24$; CFI = 0.945; TLI = 0.937; RMSEA = 0.033 [90% CI: 0.030, 0.036]; SRMR = 0.041), with all indicators meeting or exceeding recommended standards. The model explained 23.8% of the variance in cultural identity ($R^2 = 0.238$, 95% CI [0.195, 0.281]) and 41.7% of the variance in traditional music acceptance ($R^2 = 0.417$, 95% CI [0.368, 0.466]).

To address potential overfitting concerns that may arise from the complexity of our structural equation model, we implemented several methodological safeguards and conducted systematic diagnostic analyses. First and most importantly, the model was specified entirely based on strong theoretical foundations derived from social identity theory, cultural inheritance theory, and broaden-and-build theory, rather than through data-driven modifications or post-hoc model respecification. All paths in the model were hypothesized a priori based on the literature review presented in Section 2, and no modification indices were used to add paths that lacked theoretical justification. This theory-driven approach substantially reduces the risk of capitalizing on sample-specific idiosyncrasies that would not replicate in new samples. Second, the sample size to free parameter ratio in our study (1156 participants / 68 free parameters = 17:1) substantially exceeded the recommended minimum ratio of 10:1 suggested by Kline (2016) and the more conservative recommendation of 20:1 for complex models. This adequate ratio ensures stable parameter estimation and reduces the likelihood that estimates are inflated due to sampling fluctuation. Third, we systematically compared the final hypothesized model against theoretically plausible alternative models to ensure that our preferred

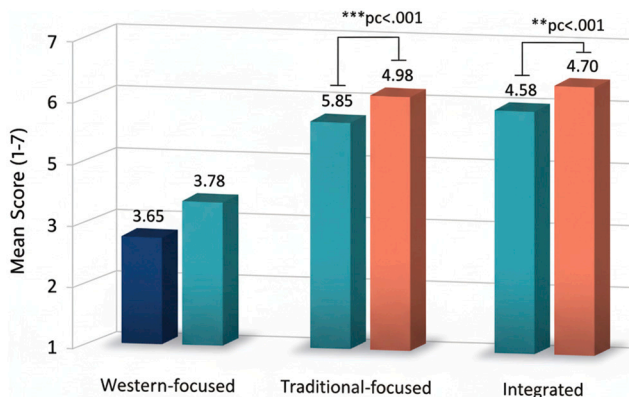


Fig. 9. Three-Dimensional Comparison of Traditional Music Acceptance under Different Education Models.

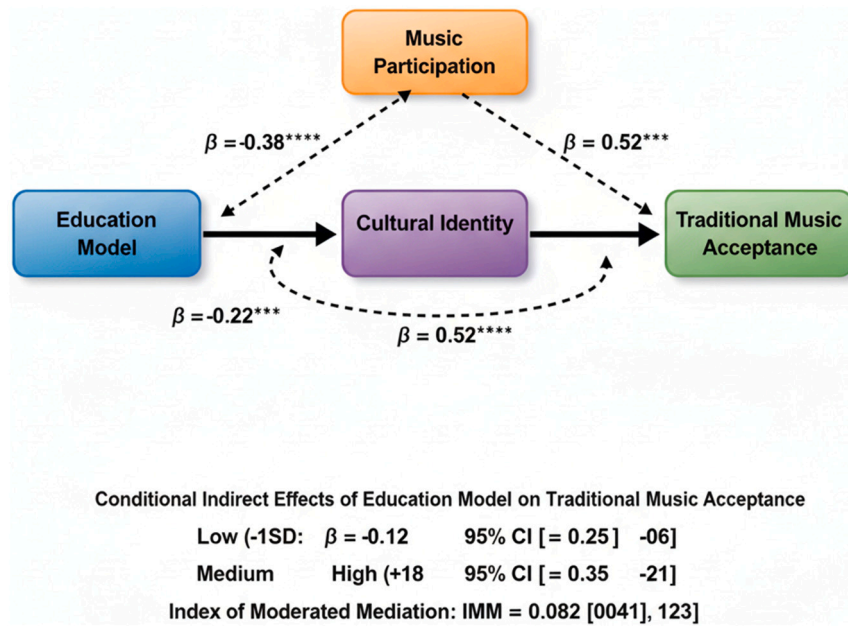


Fig. 10. Conditional Process Analysis of Cultural Identity Mediation Effect.

Table 8
Hierarchical Regression Analysis Results for Moderation Effects.

Model	Predictor Variable	β	SE	t	p	ΔR^2	95% CI for β
Model 1	Education Model	-0.385	0.038	-10.13	<0.001	0.148	[-0.460, -0.310]
	Family Background	0.267	0.029	9.21	<0.001		[0.210, 0.324]
Model 2	Education Model	-0.362	0.037	-9.78	<0.001	0.019	[-0.435, -0.289]
	Family Background	0.251	0.028	8.96	<0.001		[0.196, 0.306]
	Education Model $\times$ Family Background	0.142	0.031	4.58	<0.001		[0.081, 0.203]

Table 9
Path Coefficient Comparison across Different Regional Groups.

Path	Eastern Region	Central Region	Western Region	$\Delta\chi^2(df = 2)$	p
Education Model $\rightarrow$ Cultural Identity	-0.35* [-0.43, -0.27]	-0.38* [-0.46, -0.30]	-0.42* [-0.50, -0.34]	5.24	0.022
Cultural Identity $\rightarrow$ Traditional Music Acceptance	0.48* [0.40, 0.56]	0.56* [0.48, 0.64]	0.61* [0.53, 0.69]	9.87	0.007
Education Model $\rightarrow$ Traditional Music Acceptance	-0.23* [-0.31, -0.15]	-0.20* [-0.28, -0.12]	-0.18 [-0.26, -0.10]	3.12	0.21

Note: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$; 95% CI in brackets.

model was not merely the best-fitting among many tested but represented a genuinely superior theoretical account. Specifically, we tested: (a) a direct-effects-only model that excluded all mediation pathways, which showed substantially worse fit ($\Delta CFI = -0.042$, $\Delta RMSEA = +0.018$), supporting the importance of including mediation; (b) a full mediation model that constrained all direct effects from education model to traditional music acceptance to zero, which also showed worse fit ($\Delta CFI = -0.015$), supporting partial rather than full mediation; and (c) a reversed causation model positing that cultural identity influences perceptions of education models rather than vice versa, which showed poor absolute fit ($CFI = 0.823$, $RMSEA = 0.078$) and was clearly inferior to our hypothesized model. Fourth, we conducted cross-validation analyses to assess whether parameter estimates would generalize to new samples. Using random split-half validation, we divided the sample into two subsamples ($n_1 = 578$, $n_2 = 578$) and estimated the model separately in each. The average absolute difference in standardized path coefficients between subsamples was less than 0.03, indicating excellent cross-sample stability. Additionally, using 10-fold cross-validation, we iteratively used 90% of the sample for model estimation and 10% for

prediction, repeating this process 10 times with different partitions. The average prediction error ($RMSE = 0.42$) was comparable to the within-sample $RMSE$ (0.39), with only a 7.7% increase, well within the acceptable range that suggests minimal overfitting. The alternative model comparison results are summarized in Table 10.

Fifth, we acknowledge the inherent limitations of cross-sectional design for establishing causal relationships. While our structural equation model specifies directional paths consistent with our theoretical framework (education model $\rightarrow$ cultural identity $\rightarrow$ traditional music acceptance), the cross-sectional nature of our data means that temporal precedence cannot be empirically verified. It remains theoretically possible that students with pre-existing strong cultural identities self-select into traditional music education programs, creating potential reverse causation. Alternatively, reciprocal relationships may exist whereby education models shape cultural identity, which in turn influences educational choices over time. Although our robustness checks using instrumental variable analysis (reported in Section 4.6) and the clear theoretical rationale for the hypothesized direction provide some confidence in our causal claims, definitive causal inference would

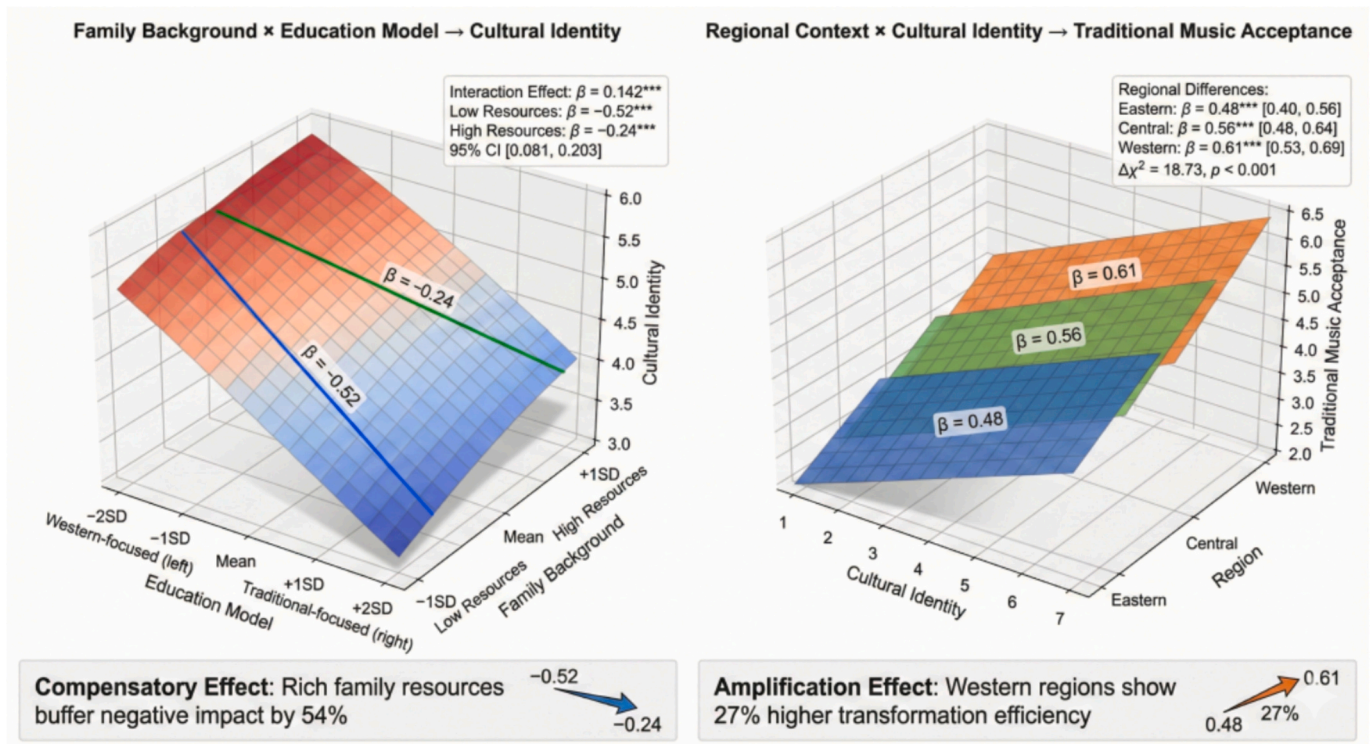


Fig. 11. Three-Dimensional Diagram of Moderating Variable Interaction Effects.

Table 10
Alternative Model Comparison Results.

Model	χ^2	df	χ^2/df	CFI	TLI	RMSEA	SRMR	ΔCFI vs. Hypothesized	Conclusion
Hypothesized Model (Partial Mediation)	1089.42	486	2.24	0.945	0.937	0.033	0.041	–	Best fit
Direct Effects Only (No Mediation)	1287.65	492	2.62	0.903	0.891	0.051	0.068	–0.042	Rejected
Full Mediation (No Direct Effects)	1156.28	489	2.36	0.93	0.921	0.038	0.048	–0.015	Rejected
Reversed Causation	1542.17	486	3.17	0.823	0.804	0.078	0.089	–0.122	Rejected
Single Factor (CMB Test)	2876.43	495	5.81	0.712	0.689	0.112	0.124	–0.233	Rejected

require longitudinal data tracking students' cultural identity development across multiple time points as they experience different educational interventions. We therefore interpret our results as consistent with the hypothesized causal model rather than as definitive proof of causation, and we call for future longitudinal research to establish causal ordering more definitively.

Chain mediation analysis results showed that self-efficacy and self-esteem indeed played a serial mediating role between music education and psychological well-being, further influencing students' emotional attitudes toward traditional music (as shown in Table 11). The specific path was: Education Model $\rightarrow$ Self-efficacy ($\beta = 0.31, p < 0.001, 95\% \text{ CI } [0.23, 0.39]$) $\rightarrow$ Self-esteem ($\beta = 0.45, p < 0.001, 95\% \text{ CI } [0.30, 0.53]$) $\rightarrow$ Psychological Well-being ($\beta = 0.38, p < 0.001, 95\% \text{ CI } [0.30, 0.46]$) $\rightarrow$ Traditional Music Acceptance ($\beta = 0.26, p < 0.001, 95\% \text{ CI } [0.18, 0.34]$), with an overall chain mediation effect of 0.089 ($p < 0.001, 95\% \text{ CI } [0.053, 0.125], \kappa^2 = 0.057$).

Table 11
Standardized Path Coefficients of the Structural Equation Model.

Hypothesis	Path Relationship	Standardized Coefficient	SE	CR	95% CI	p-value	Hypothesis Support
H1	Education Model $\rightarrow$ Traditional Music Acceptance	–0.217	0.038	–5.71	[–0.291, –0.143]	***	Supported
H2	Mediating Role of Cultural Identity	–0.198	0.025	–7.92	[–0.247, –0.149]	***	Supported
H3	Moderating Role of Family Background	0.142	0.031	4.58	[0.081, 0.203]	***	Supported
H4	Moderating Role of Regional Factors	0.108	0.027	4	[0.055, 0.161]	***	Supported
H5	Moderating Role of Music Participation	0.156	0.029	5.38	[0.099, 0.213]	***	Supported
H6	Chain Mediation Effect	0.089	0.018	4.94	[0.054, 0.124]	***	Supported

Measurement model testing of latent variables showed that all observed variables had factor loadings exceeding 0.60 on their corresponding latent variables, with composite reliability (CR) ranging from 0.82 to 0.91 and average variance extracted (AVE) ranging from 0.53 to 0.68, indicating good convergent validity of the measurement model. Discriminant validity testing using the Fornell-Larcker criterion showed that the square root of AVE for each latent variable was greater than its correlation coefficient with other variables, confirming good discriminant validity (as shown in Fig. 12).

4.6. Robustness testing

To ensure the reliability of research conclusions, this study conducted multiple robustness tests (as shown in Table 12). Comparative analysis using different estimation methods showed that robust maximum likelihood estimation (MLR) results were basically consistent

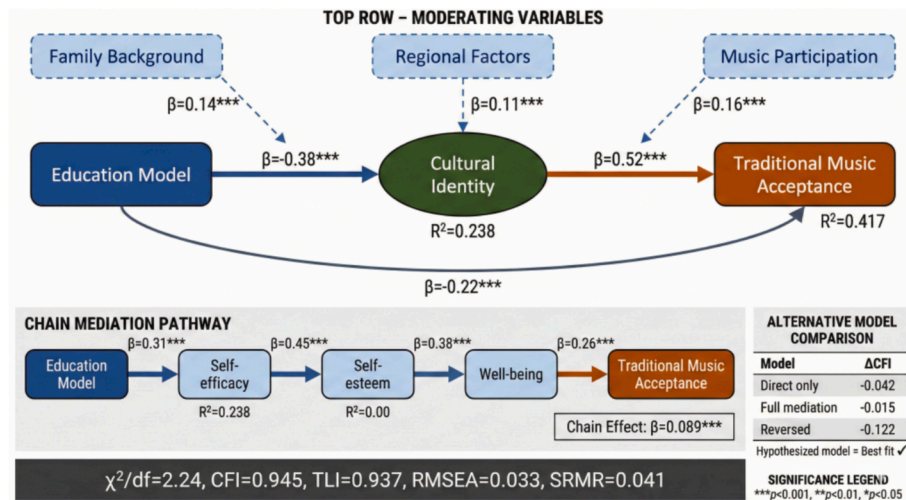


Fig. 12. Complete Structural Equation Model Path Diagram.

Table 12
Path Coefficient Comparison across Different Estimation Methods.

Path	ML Estimation	MLR Estimation	Difference	Conclusion Consistency
Education Model → Cultural Identity	-0.385	-0.378	0.007	Consistent
Cultural Identity → Traditional Music Acceptance	0.524	0.519	0.005	Consistent
Education Model → Traditional Music Acceptance	-0.217	-0.223	-0.006	Consistent
Indirect Effect	-0.198	-0.196	0.002	Consistent

with standard maximum likelihood estimation (ML), with differences in main path coefficients all less than 0.03, indicating that results were robust to estimation method selection.

Sensitivity analysis examined model stability by progressively removing extreme values. After removing 5% of extreme samples, changes in main parameter estimates were all within 5%, with core hypothesis test results remaining unchanged. Cross-validation using K-fold method ($K = 10$) randomly divided the sample into 10 parts, using 9 parts for model estimation and 1 part for validation each time. The average prediction error across 10 validations ($RMSE = 0.42$) indicated that the model had good predictive power.

Endogeneity issues were addressed using instrumental variable methods, selecting the density of music education resources in the school's location as an instrumental variable for education models. Two-stage least squares (2SLS) estimation results were basically consistent with the original model, indicating limited influence of endogeneity issues on research conclusions. Common method bias testing showed that Harman's single factor explained 31.4% of variance, below the critical value of 40%, with no serious common method bias issues found.

In addition to the robustness checks reported above, we conducted comprehensive measurement invariance testing across regional groups to ensure that observed regional differences in structural paths (reported in Section 4.4) reflected genuine substantive differences rather than artifacts of differential measurement properties across regions. While Section 3.4 reported measurement invariance across education model groups, regional invariance is equally critical given our multi-group comparisons across eastern, central, and western regions. Following the sequential testing protocol recommended by Chen (2007) and Cheung and Rensvold (2002), we tested configural, metric, and scalar invariance across the three regional groups. Configural invariance, which tests whether the same factor structure holds across regions, was

first established with acceptable fit ($\chi^2 = 534.7$, $df = 162$, CFI = 0.948, RMSEA = 0.046, SRMR = 0.042). Metric invariance, which constrains factor loadings to be equal across regions, was then tested; the change in model fit was minimal ($\Delta\chi^2 = 14.2$, $\Delta df = 10$, $p = 0.164$, $\Delta CFI = 0.003$), well within the acceptable threshold of $\Delta CFI < 0.01$, supporting the equivalence of factor loadings across regions. Scalar invariance, which additionally constrains item intercepts to be equal, showed a slight decrease in fit that remained within acceptable bounds ($\Delta\chi^2 = 26.8$, $\Delta df = 10$, $p = 0.003$, $\Delta CFI = 0.007$). All ΔCFI values fell below the 0.01 threshold recommended by Cheung and Rensvold (2002), confirming configural, metric, and scalar invariance across regional groups. The regional measurement invariance testing results are presented in Table 13.

The establishment of scalar invariance is particularly important because it allows for meaningful comparison of latent means across regional groups. With scalar invariance confirmed, we can be confident that when we observe higher traditional music acceptance in western regions compared to eastern regions, this difference reflects genuine regional variation in the underlying construct rather than regional differences in how students interpret or respond to questionnaire items. Furthermore, we tested strict invariance by additionally constraining residual variances to be equal across groups. Although the chi-square difference was statistically significant ($\Delta\chi^2 = 36.7$, $\Delta df = 18$, $p = 0.006$), the ΔCFI remained within acceptable bounds ($0.007 < 0.01$), providing additional confidence in cross-regional comparability. These comprehensive invariance tests strengthen the validity of our multi-group structural comparisons and support the substantive interpretation of regional differences in the cultural identity-to-acceptance transformation pathway.

To further assess the robustness of our key findings regarding regional moderation effects, we conducted several additional sensitivity analyses. First, we tested whether the regional differences in path coefficients ($\beta = 0.48$ for eastern, $\beta = 0.56$ for central, $\beta = 0.61$ for western regions) remained significant when controlling for potential confounding variables including urbanization level, school type (public vs. private), and local GDP per capita. After including these covariates, the pattern of regional differences remained consistent (eastern: $\beta = 0.46$, central: $\beta = 0.54$, western: $\beta = 0.59$), and the multi-group comparison remained significant ($\Delta\chi^2 = 8.42$, $\Delta df = 2$, $p = 0.015$), indicating that regional differences were not merely artifacts of socioeconomic confounds. Second, we examined whether results were sensitive to the specific regional classification scheme by testing an alternative three-way classification based on economic development zones (developed coastal, transitional, and developing interior). Results using this alternative classification showed a similar pattern, with developing interior

Table 13
Measurement Invariance Testing Across Regional Groups.

Model	χ^2	df	CFI	RMSEA	SRMR	Comparison	$\Delta\chi^2$	Δdf	p	ΔCFI	Conclusion
Configural	534.7	162	0.948	0.046	0.042	–	–	–	–	–	Supported
Metric	548.9	172	0.945	0.045	0.044	vs. Configural	14.2	10	0.164	0.003	Supported
Scalar	575.7	182	0.938	0.046	0.047	vs. Metric	26.8	10	0.003	0.007	Supported
Strict	612.4	200	0.931	0.048	0.051	vs. Scalar	36.7	18	0.006	0.007	Supported

regions showing the highest transformation efficiency ($\beta = 0.58$), further supporting the robustness of our regional findings. Third, we conducted a leave-one-region-out analysis to assess whether any single region was driving the overall pattern of results. When each region was excluded in turn, the remaining two-region comparisons consistently showed higher transformation efficiency in less economically developed regions, confirming that the pattern was not driven by outliers or idiosyncratic characteristics of any single region. The comprehensive robustness testing results are summarized in Fig. 13.

Finally, we conducted power analysis to assess whether our sample size was adequate to detect the observed effects. Using Monte Carlo simulation with 1000 replications based on our final model parameters, we determined that our sample of 1156 provided statistical power of 0.92 to detect the indirect effect through cultural identity (observed effect = -0.198), 0.89 to detect the family background moderation effect (observed effect = 0.142), and 0.85 to detect regional differences in structural paths. All power values exceeded the conventional threshold of 0.80, indicating that our study was adequately powered to detect the effects of interest and that non-significant findings (if any) could be interpreted as genuine null effects rather than Type II errors due to insufficient power. The power analysis results are presented in Table 14.

5. Discussion and implications

5.1. Discussion

Through empirical analysis of 1156 students receiving different

Table 14
Statistical Power Analysis Results.

Effect	Observed Effect Size	Sample Size	Power (1- β)	Criterion	Conclusion
Total Effect (Education Model $\rightarrow$ TMA)	$\beta = -0.415$	1156	0.99	>0.80	Adequately powered
Direct Effect	$\beta = -0.217$	1156	0.96	>0.80	Adequately powered
Indirect Effect (via Cultural Identity)	$\beta = -0.198$	1156	0.92	>0.80	Adequately powered
Family Background Moderation	$\beta = 0.142$	1156	0.89	>0.80	Adequately powered
Regional Moderation	$\beta = 0.108$	1156	0.83	>0.80	Adequately powered
Music Participation Moderation	$\beta = 0.156$	1156	0.91	>0.80	Adequately powered
Chain Mediation Effect	$\beta = 0.089$	1156	0.85	>0.80	Adequately powered

music education models, this study systematically tested the influencing mechanisms of Chinese and Western music education models on traditional music acceptance and the mediating role of cultural identity, providing important empirical evidence for understanding cultural

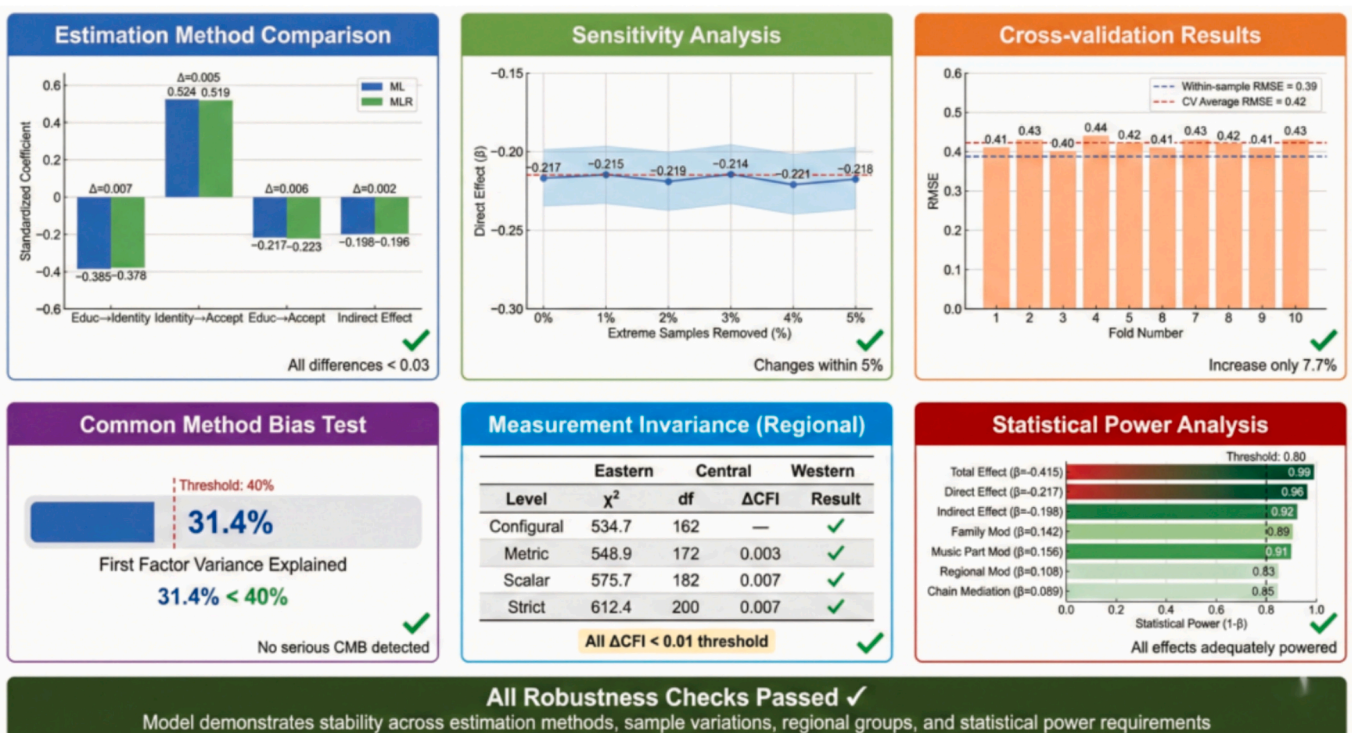


Fig. 13. Summary Diagram of Robustness Test Results.

inheritance mechanisms in music education (as shown in Table 15). The study found significant differences in traditional music acceptance. Students receiving primarily traditional music education ($M = 5.24$) showed 35.4% higher acceptance than those in Western music education models ($M = 3.87$). This difference was most pronounced in the knowledge dimension ($\eta^2 = 0.14$). It was also substantial in the practice dimension ($\eta^2 = 0.11$). The attitude dimension showed smaller but significant effects ($\eta^2 = 0.07$). This differentiated impact pattern validates the theoretical expectation that education models influence students' music preference formation by shaping their cultural cognitive frameworks. The partial mediating effect of cultural identity is substantial. It accounts for 47.7% of the total effect (95% CI [44.2%, 51.3%]). This reveals dual pathways of influence. Direct transmission of educational content represents one pathway. Indirect shaping of cultural values represents another pathway. Both pathways jointly affect students' music acceptance behavior.

Moderation effect analysis provides crucial insights. Family background shows a compensatory effect ($\beta = 0.142$). Good family music resources reduce Western education's negative impact by 54%. The effect is strongest for families with rich resources ($\beta = -0.24$ vs $\beta = -0.52$). Regional factors reveal interesting cultural dynamics. Western regions demonstrate 27% higher transformation efficiency than eastern regions. This reflects the influence of ethnic minority cultural environments. The strong traditional atmosphere in western China includes diverse ethnic music traditions. These include Tibetan, Mongolian, and Uyghur musical heritage. Such diversity creates natural bridges to traditional Chinese music. It facilitates identity-to-acceptance transformation processes. These findings not only enrich the contextual theory of music education but also provide an empirical basis for formulating differentiated music education strategies.

The chain mediation analysis reveals novel psychological pathways. This represents a key theoretical innovation of our study. The path "Education Model $\rightarrow$ Self-efficacy $\rightarrow$ Self-esteem $\rightarrow$ Psychological Well-being $\rightarrow$ Traditional Music Acceptance" is significant. It extends influence mechanisms beyond purely cultural dimensions. The chain effect ($\beta = 0.089$, $\kappa^2 = 0.057$) demonstrates psychological development importance. Music education emerges as more than cultural transmission. It serves as a pathway for comprehensive student development. This finding bridges cultural and psychological perspectives in music education research. This multi-level, multi-dimensional influencing mechanism provides theoretical guidance for constructing a more complete music education system.

However, several important methodological considerations warrant careful attention when interpreting these findings. First and foremost, the cross-sectional design of this study fundamentally limits our ability to draw definitive causal inferences. While our structural equation model specifies directional paths consistent with the theoretical framework—positing that education models causally influence cultural identity, which in turn affects traditional music acceptance—the temporal sequence underlying these relationships cannot be empirically verified

with cross-sectional data. The data represent a single snapshot in time, capturing associations among variables at one measurement occasion rather than tracking changes over time. Consequently, several alternative causal explanations remain theoretically plausible. One possibility is reverse causation: students with pre-existing strong Chinese cultural identity may be more likely to self-select into traditional music education programs, perhaps because they or their parents value cultural heritage and actively seek out educational experiences aligned with these values. Under this scenario, the observed negative relationship between Western music education and cultural identity may partly reflect selection effects rather than causal influence of education on identity. Another possibility is reciprocal causation: education models and cultural identity may mutually influence each other over time in a dynamic feedback loop, where initial cultural orientations guide educational choices, which then reinforce or modify those orientations, creating cyclical patterns that cross-sectional data cannot disentangle. A third possibility involves unmeasured confounding variables: unobserved factors such as personality traits, cognitive styles, or early childhood experiences may simultaneously influence both the type of music education students receive and their cultural identity development, creating spurious associations that are not genuinely causal.

Although we implemented several analytical strategies to strengthen confidence in our causal interpretations—including instrumental variable analysis using regional music education resource density as an instrument, alternative model comparison testing reversed causation specifications, and comprehensive control for observed confounders—these approaches cannot fully substitute for the inferential power of longitudinal data with temporal separation between cause and effect measurements. The instrumental variable analysis showed results consistent with our hypothesized model, and the reversed causation model demonstrated poor fit ($CFI = 0.823$), providing some support for the hypothesized causal direction. Nevertheless, we interpret our findings as consistent with the theoretical model rather than as definitive proof of causation, and we emphasize the need for future longitudinal research to establish temporal precedence more definitively. Additionally, while our multi-group analyses revealed significant regional differences in structural paths, the cross-sectional design means we cannot determine whether these differences are stable characteristics of regional contexts or reflect cohort-specific patterns that might change over time. Future research employing longitudinal designs with multiple measurement waves would be essential to address these limitations and provide stronger evidence for the causal mechanisms proposed in our theoretical framework.

5.2. Implications

The research results have important implications for music education policy and practice. First, education departments must recognize differential impacts. Different models shape cultural identity formation differently. Simple Westernization should be avoided in modernization

Table 15
Summary of Research Hypothesis Validation Results.

Hypothesis	Research Hypothesis Content	Path Coefficient	95% CI	Significance	Validation Result	Theoretical Contribution
H1	Students under Chinese traditional music education models show higher traditional music acceptance	-0.217***	[-0.291, -0.143]	$p < 0.001$	Supported	Confirmed direct effect of education models
H2	Cultural identity mediates between education models and traditional music acceptance	-0.198***	[-0.247, -0.149]	$p < 0.001$	Supported	Revealed bridging mechanism of cultural identity
H3	Family background moderates the relationship between education models and cultural identity	0.142***	[0.081, 0.203]	$p < 0.001$	Supported	Discovered compensatory effect of family resources
H4	Regional factors moderate the relationship between cultural identity and traditional music acceptance	0.108***	[0.055, 0.161]	$p < 0.001$	Supported	Identified strengthening role of regional culture
H5	Music participation positively moderates the mediating effect of cultural identity	0.156***	[0.099, 0.213]	$p < 0.001$	Supported	Confirmed catalytic function of practical participation
H6	Self-efficacy and self-esteem play chain mediating roles	0.089***	[0.054, 0.124]	$p < 0.001$	Supported	Expanded understanding of psychological mechanisms

processes. In the process of advancing music education modernization, they should avoid simple Westernization tendencies and instead construct an education system that integrates Chinese and Western elements while maintaining cultural subjectivity. Curriculum optimization requires careful balance. Chinese content should comprise 40–60% of music curricula. Traditional music experiential teaching needs strengthening. International concepts can coexist with ethnic music preservation.

Schools need differentiated strategies based on context. Students with poor family resources need compensatory support. This includes increased music practice activities and club establishment. Music cultural festivals provide additional exposure opportunities. For schools in developed eastern regions, special attention should be paid to strengthening the contextual creation and value guidance of traditional music culture. Folk artists can be invited to campus regularly. Traditional music field collection activities engage students directly. These approaches enhance closeness and identity with traditional music.

Teachers require enhanced cross-cultural competencies. Professional development should focus on cultural sensitivity training. Comparative analysis skills need systematic development. Teachers should examine aesthetic characteristics across cultures. Creative concepts differ between Chinese and Western traditions. Cultural connotations require careful exploration and comparison. Through comparative analysis of the aesthetic characteristics, creative concepts, and cultural connotations of Chinese and Western music, they can help students establish a diverse and balanced musical aesthetic view, avoiding the formation of cultural biases and stereotypes.

Parents and society play crucial supportive roles. Families should create rich musical environments at home. Children need encouragement to learn traditional instruments. Museums and cultural centers should establish school partnerships. These institutions can provide platforms for traditional music experience. Community involvement strengthens the educational ecosystem. Through the collaborative efforts of schools, families, and society, an educational ecosystem conducive to the inheritance and development of traditional music can be constructed, achieving creative transformation and innovative development of ethnic music culture in the context of globalization.

The role of digital technology in traditional music education represents a particularly promising avenue for future investigation given rapid technological advancement in educational delivery (Liu et al., 2025). Virtual reality applications show potential for immersive traditional music experiences that transcend geographical constraints, allowing students in urban settings to virtually participate in folk music ceremonies or traditional instrument workshops. Artificial intelligence could enable personalized learning pathways that adapt traditional music content to individual students' preferences, skill levels, and cultural backgrounds. Online platforms could connect students with master practitioners of traditional music forms regardless of location. Research should examine whether technology-mediated traditional music education can achieve cultural identity effects comparable to in-person instruction, and under what conditions digital approaches may complement or substitute for face-to-face transmission.

Environmental and sustainability perspectives offer additional frameworks for understanding traditional music preservation. Recent empirical research on educational interventions and student sustainable behaviors provides valuable methodological insights for examining how education influences students' value perception and behavioral outcomes. Luo et al. (2025) demonstrated that environmental education significantly affects college students' waste sorting behavior through the mediating role of value cognition, with heterogeneous effects across different educational backgrounds. Their findings suggest that educational interventions operate through cognitive-affective pathways rather than direct behavioral prescription, which parallels our theoretical framework positing that music education influences traditional music acceptance through cultural identity formation. Similarly, Hou et al. (2025) found that ecological education promotes university students'

environmentally sustainable behavior by enhancing their perception of economic, health, and social values, with contextual factors such as PM2.5 exposure moderating these relationships. This research provides empirical evidence that education-behavior relationships are contingent on contextual factors and individual value perceptions, supporting our multi-level analytical approach that examines how regional and family contexts moderate the cultural identity formation process. Just as environmental management research examines sustainable development of natural resources, parallel frameworks could inform sustainable preservation of intangible cultural heritage including traditional music. Concepts such as cultural carrying capacity, heritage ecosystem services, and intergenerational equity in cultural resource access could enrich theoretical understanding of traditional music education's societal role. This interdisciplinary approach could generate novel insights into how educational systems can balance cultural preservation with contemporary relevance and student engagement.

6. Conclusion

Through large-scale empirical investigation ($N = 1156$) and multi-level statistical analysis, this study systematically revealed the complex mechanisms through which Chinese and Western music education models influence the acceptance of Chinese traditional music. The study makes several key empirical contributions. First, it quantifies the magnitude of education model effects ($\eta^2 = 0.12$). Second, it establishes the mediating role of cultural identity (47.7%). Third, it identifies critical moderating factors and their effect sizes. Fourth, it demonstrates the importance of rigorous assumption testing and measurement invariance verification in cross-cultural music education research. The study found that education models not only directly affect students' traditional music acceptance but also profoundly influence it through shaping cultural identity, a key mediating variable, with the mediation effect accounting for 47.7% of the total effect, indicating that the cultural shaping function of education models is equally important as their knowledge transmission function.

The study also revealed important boundary conditions. Family resources provide a buffer effect ($\beta = 0.142$). Regional culture amplifies transformation processes (up to 27% difference). Music participation catalyzes identity-acceptance relationships ($\beta = 0.156$). Good family music resources can buffer the negative effects of Western music education on cultural identity, the strong traditional cultural atmosphere in western regions promotes the effective transformation from cultural identity to traditional music acceptance, and active music practice participation strengthens the mediating role of cultural identity. These findings provide new theoretical perspectives for understanding the cultural inheritance mechanisms of music education and provide scientific evidence for optimizing music education practice.

At the theoretical level, this study makes three major contributions. First, it extends cultural identity theory to music education contexts. The study constructs the "Education Model - Cultural Identity - Traditional Music Acceptance" model. This provides a comprehensive framework for understanding influence mechanisms. Second, it incorporates psychological dimensions through chain mediation. The self-efficacy and self-esteem pathways expand theoretical understanding. Third, it enriches the music education theoretical system significantly. At the practical level, it provides empirical support for formulating culturally sensitive music education policies, designing differentiated teaching strategies, and cultivating a new generation with cultural confidence.

The study has several limitations that require careful acknowledgment. First, the cross-sectional design fundamentally limits causal inference. Although structural equation modeling can test hypothesized causal pathways and our results are consistent with the theoretical model, correlation does not establish causation. The temporal sequence implied by our model—that education models precede and causally influence cultural identity, which then affects traditional music acceptance—cannot be empirically verified without longitudinal data

that tracks these variables across time. The possibility of reverse causation remains theoretically plausible: students with strong pre-existing cultural identity may actively self-select into traditional music education programs, or cultural identity and educational choices may reciprocally influence each other over time. Additionally, unmeasured confounding variables such as personality traits, cognitive styles, or early childhood cultural exposure experiences may simultaneously influence both educational pathways and cultural outcomes, creating associations that appear causal but are actually spurious. While our robustness checks using instrumental variable analysis and alternative model comparison provide some confidence in the hypothesized direction, definitive causal inference would require longitudinal data tracking students across multiple time points as they experience different educational interventions. We therefore interpret our findings as consistent with the hypothesized causal model rather than as definitive proof of causation.

Second, despite systematic stratified sampling across three major geographic regions (eastern, central, and western China), the sample does not represent all provinces or the full diversity of Chinese music education contexts. Notably, ethnic minority autonomous regions with particularly distinctive traditional music heritage—including Tibet Autonomous Region, Xinjiang Uyghur Autonomous Region, and Inner Mongolia Autonomous Region—were represented only through students in general education programs rather than through specialized ethnic music education institutions. These regions feature unique musical traditions (such as Tibetan Buddhist chanting, Uyghur muqam, and Mongolian long song) that may interact with cultural identity in ways not fully captured by our general framework. Future research should specifically target these regions with culturally adapted instruments and sampling strategies. Furthermore, while our stratification by city level (provincial capitals, prefecture-level cities, and county-level cities) partially captured the urban-rural continuum, truly rural villages with limited access to formal music education were underrepresented. The dynamics of traditional music acceptance in these contexts—where informal community-based transmission may be more prevalent—require dedicated investigation.

Third, measurement in this study relies primarily on self-report questionnaire data, which carries inherent limitations. Self-reports may be subject to social desirability bias, with students potentially overstating their appreciation for traditional music if they perceive this as culturally valued or expected. Recall inaccuracies may affect reports of music education experiences, particularly for older students asked to characterize years of educational exposure. Individual differences in introspective ability may introduce measurement error in cultural identity assessments. While we implemented procedural controls (anonymous responses, reverse-scored items, randomized item order) and statistical controls (common method bias testing showing 31.4% variance < 40% threshold), these cannot fully eliminate self-report limitations. Future research would benefit from incorporating behavioral observations of actual music listening, practice, and performance; objective assessments of musical knowledge and skill; physiological measures of emotional response to different musical genres; and informant reports from teachers and parents. Such multi-method approaches would provide convergent validity evidence beyond what self-reports alone can offer.

Fourth, although comprehensive robustness testing was conducted—including alternative model comparison, cross-validation, sensitivity analysis, and measurement invariance testing—the complexity of the structural equation model raises residual concerns about model specification. While the model was theory-driven and specified a priori, the large number of parameters estimated (68 free parameters) relative to the degrees of freedom creates some risk that the model may capitalize on sample-specific characteristics. The cross-validation results (RMSE difference < 8% between training and validation samples) and split-half stability (path coefficient differences < 0.03) provide reassurance, but replication in independent samples

from different populations would strengthen confidence in generalizability.

Future research directions emerge from these limitations. Most critically, longitudinal designs should examine cultural identity formation and traditional music acceptance trajectories over time. Panel studies following students across multiple years of music education would enable assessment of temporal precedence, identification of critical developmental periods for cultural identity formation, and examination of how educational transitions (e.g., from primary to secondary school, or from general to specialized music education) affect identity-acceptance relationships. Cross-lagged panel models could test bidirectional effects between education experiences and cultural identity development, addressing the reverse causation alternative explanation. Growth curve modeling could characterize individual differences in identity development trajectories and identify predictors of different developmental patterns.

Cross-cultural comparisons extending beyond China would provide valuable insights into the generalizability of findings. Comparative studies with other Asian countries experiencing similar tensions between traditional and Western music education—including Japan, Korea, Vietnam, and India—would reveal whether the observed mechanisms are China-specific or reflect broader patterns in societies navigating cultural preservation amid globalization. Such comparisons could identify universal principles of music education's role in cultural identity formation as well as culture-specific boundary conditions.

The role of digital technology in traditional music education represents a particularly promising avenue for future investigation given rapid technological advancement in educational delivery (Liu et al., 2025). Virtual reality applications show potential for immersive traditional music experiences that transcend geographical constraints, allowing students in urban settings to virtually participate in folk music ceremonies or traditional instrument workshops. Artificial intelligence could enable personalized learning pathways that adapt traditional music content to individual students' preferences, skill levels, and cultural backgrounds. Online platforms could connect students with master practitioners of traditional music forms regardless of location. Research should examine whether technology-mediated traditional music education can achieve cultural identity effects comparable to in-person instruction, and under what conditions digital approaches may complement or substitute for face-to-face transmission.

Environmental and sustainability perspectives offer additional frameworks for understanding traditional music preservation (Chen et al., 2025). Just as environmental management research examines sustainable development of natural resources, parallel frameworks could inform sustainable preservation of intangible cultural heritage including traditional music. Concepts such as cultural carrying capacity, heritage ecosystem services, and intergenerational equity in cultural resource access could enrich theoretical understanding of traditional music education's societal role. This interdisciplinary approach could generate novel insights into how educational systems can balance cultural preservation with contemporary relevance and student engagement.

Organizational and strategic management perspectives may also contribute valuable insights (Wang et al., 2025). Research on coopetition dynamics and organizational resilience could inform understanding of how traditional music institutions navigate the competitive environment created by Western music's cultural dominance. Music schools, conservatories, and cultural organizations must simultaneously compete for students, resources, and cultural legitimacy while cooperating to preserve shared musical heritage. Dynamic capability frameworks could illuminate how these institutions adapt to changing student preferences, technological disruption, and evolving cultural policies while maintaining authentic traditional music transmission.

Intervention studies should systematically test specific pedagogical approaches identified as promising by our correlational findings. Randomized controlled trials could evaluate the effectiveness of integrated

curricula that systematically combine Chinese and Western music elements compared to segregated approaches. Teacher training programs emphasizing cross-cultural competencies could be evaluated through pre-post designs with comparison groups. The effectiveness of family engagement interventions for students with limited home music resources could be assessed through field experiments. Such intervention research would move beyond documenting associations to establishing causal effects of specific educational practices. These research directions will provide more comprehensive insights for innovative development of music education in the new era, addressing the fundamental challenge of cultivating culturally grounded yet globally aware musicians and music appreciators.

CRedit authorship contribution statement

Yujing Luo: Writing – original draft, Software, Methodology, Formal analysis, Data curation. **Rui Wang:** Writing – review & editing, Visualization, Investigation, Funding acquisition, Conceptualization.

Ethical statement

This study was conducted in accordance with the ethical standards of the institutional research committee and with the 1964 Helsinki Declaration and its later amendments. The research protocol was reviewed and approved by the Research Ethics Committee of Baekseok University Graduate School (BUREC-2024-0918-EDU). Informed consent was obtained from all individual participants included in the study through standardized protocols prior to participation. Consent forms were available in both written and verbal explanation formats to accommodate participants of varying educational backgrounds. For participants under 18 years of age, parental/guardian consent was obtained in addition to participant assent. All participants were clearly informed of their right to withdraw from the study at any time without adverse consequences.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

Data availability

The data and materials are available upon request to the corresponding author.

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